

MTH251 & MTH252

Mathematical Analysis I & II

Course Textbook

Department of Mathematical Sciences
UNIST

Mathematical Analysis I & II

Course Textbook

Author

MUHO CHOI (*Undergraduate Student of CSE, UNIST*)

Email: sihyeon8240@unist.ac.kr

GitHub: <https://github.com/sihyeon8240>

Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/Math>

Feel free to open issues or pull requests for corrections.

© 2026. MUHO CHOI. All rights reserved.

Developed for the Department of Mathematical Sciences, UNIST.

This textbook is licensed under a **Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License** (CC BY-NC-SA 4.0).

You are free to: Share, copy, and redistribute the material in any medium or format; adapt, remix, transform, and build upon the material.

Under the following terms: You must give appropriate credit; you may not use the material for commercial purposes; if you remix or transform the material, you must distribute your contributions under the same license.

Contents

1	The Number Systems	1
1.1	The Natural, Integer, and Rational Number Systems	1
1.2	Ordered Sets	3
1.3	Fields and Ordered Fields	5
1.4	The Real Field	6
1.5	The Extended Real Number System	15
1.6	The Complex Field	16
1.7	Euclidean Spaces	19
2	Basic Topology	21
2.1	Relations	21
2.2	Functions	22
2.3	Finite Sets	25
2.4	Countable and Uncountable Sets	34
2.5	Metric Spaces	43
2.6	Compact Sets	53
2.7	Connected Sets	63
3	Numerical Sequences and Series	65
3.1	Convergent Sequences	65
3.2	Subsequences	65
3.3	Cauchy Sequences	65
3.4	Upper and Lower Limits	65
3.5	Some Special Sequences	65
3.6	Series	65
3.7	Series of Nonnegative Terms	65
3.8	The Number e	65
3.9	The Root and Ratio Tests	65
3.10	Power Series	65
3.11	Summation by Parts	66
3.12	Absolute Convergence	66
3.13	Addition and Multiplication of Series	66
3.14	Rearrangements	66

4	Continuity	67
4.1	Limits of Functions	67
4.2	Continuous Functions	67
4.3	Continuity and Compactness	67
4.4	Continuity and Connectedness	67
4.5	Discontinuities	67
4.6	Monotonic Functions	67
4.7	Infinite Limits and Limits at Infinity	67
5	Differentiation	68
5.1	The Derivative of a Real Function	68
5.2	Mean Value Theorems	68
5.3	The Continuity of Derivatives	68
5.4	L'Hospital's Rule	68
5.5	Derivatives of Higher Order	68
5.6	Taylor's Theorem	68
5.7	Differentiation of Vector-valued Functions	68
6	The Riemann-Stieltjes Integral	69
6.1	Definition and Existence of the Integral	69
6.2	Properties of the Integral	69
6.3	Integration and Differentiation	69
6.4	Integration of Vector-valued Functions	69
6.5	Rectifiable Curves	69
7	Sequences and Series of Functions	70
7.1	Discussion of Main Problem	70
7.2	Uniform Convergence	70
7.3	Uniform Convergence and Continuity	70
7.4	Uniform Convergence and Integration	70
7.5	Uniform Convergence and Differentiation	70
7.6	Equicontinuous Families of Functions	70
7.7	Equicontinuous Families of Functions	70
7.8	The Stone-Weierstrass Theorem	70
8	Some Special Functions	71
8.1	Power Series	71
8.2	The Exponential and Logarithmic Functions	71
8.3	The Trigonometric Functions	71
8.4	The Algebraic Completeness of the Complex Field	71
8.5	Fourier Series	71
8.6	The Gamma Function	71

9	Functions of Several Variables	72
9.1	Linear Transformations	72
9.2	Differentiation	72
9.3	The Contraction Principle	72
9.4	The Inverse Function Theorem	72
9.5	The Implicit Function Theorem	72
9.6	The Rank Theorem	72
9.7	Determinants	72
9.8	Derivatives of Higher Order	72
9.9	Differentiation of Integrals	72
10	Integration of Differential Forms	73
10.1	Integration	73
10.2	Primitive Mappings	73
10.3	Partitions of Unity	73
10.4	Change of Variables	73
10.5	Differential Forms	73
10.6	Simplexes and Chains	73
10.7	Stokes' Theorem	73
10.8	Closed Forms and Exact Forms	73
10.9	Vector Analysis	73
11	The Lebesgue Theory	74
11.1	Set Functions	74
11.2	Construction of the Lebesgue Measure	74
11.3	Measure Spaces	74
11.4	Measurable Functions	74
11.5	Simple Functions	74
11.6	Integration	74
11.7	Comparison with the Riemann Integral	74
11.8	Integration of Complex Functions	74
11.9	Functions of Class $\mathcal{L}^2$	74
	Appendix: Construction of $\mathbb{R}$	75

Chapter 1

The Number Systems

1.1 The Natural, Integer, and Rational Number Systems

We denote the set $\{1, 2, 3, \dots\}$ of natural numbers by $\mathbb{N}$. The set $\mathbb{N}$ is defined by the following axioms:

- $1 \in \mathbb{N}$.
- If $n \in \mathbb{N}$, then $s(n) \in \mathbb{N}$ where $s(n) = n + 1$ is called the **successor** of n .
- 1 is not the successor of any element of $\mathbb{N}$.
- If $m, n \in \mathbb{N}$ and $s(m) = s(n)$, then $m = n$.
- Any subset of $\mathbb{N}$ which contains 1, and contains $s(n)$ whenever it contains n , must be equal to $\mathbb{N}$.

The axioms above are called the **Peano axioms** (or *Peano Postulates*).

Remark. The **Peano axioms** does not guarantee the *existence* of $\mathbb{N}$, which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** will not be discussed, so we will just assume that $\mathbb{N}$ exists. Also, we will reject the **axiom of choice** and provide constructive proofs as far as possible.

Remark. Let $P(n)$ be a statement about $n \in \mathbb{N}$, and let $S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\}$. Assume that $P(n)$ satisfies the following conditions:

- $P(1)$ is true.
- $P(n + 1)$ is true whenever $P(n)$ is true.

Then, we have the following:

- $1 \in S$.

- $n + 1 \in S$ whenever $n \in S$.

Therefore, we have $S = \mathbb{N}$, i.e., $P(n)$ is true for all $n \in \mathbb{N}$. This method of proof is called the **principle of mathematical induction**.

Theorem 1.1.1. *Any nonempty subset of $\mathbb{N}$ has a least element. This theorem is called the **well-ordering property** of $\mathbb{N}$.*

Proof. Let S be a nonempty subset of $\mathbb{N}$. Assume that S does not have a least element. Let T and L be defined as follows:

- $T := \mathbb{N} \setminus S$.
- $L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\}$.

We will show that $L = \mathbb{N}$, which implies that $S = \emptyset$, a contradiction.

- **Base case:** $1 \in L$ since $\{1\} \subseteq T$. Otherwise, $1 \in S$ and 1 is the least element of S .
- **Inductive step:** Assume that $n \in L \implies \{1, 2, \dots, n\} \subseteq T$. Then $n + 1 \notin S$. Otherwise, $n + 1$ is the least element of S . Therefore, we have

$$n + 1 \in T \implies \{1, 2, \dots, n + 1\} \subseteq T.$$

Thus $n + 1 \in L$.

By the principle of mathematical induction, we have $L = \mathbb{N}$. □

We define the set $\mathbb{Z}$ of integers and the set $\mathbb{Q}$ of rational numbers as follows:

- $\mathbb{Z} := \{m - n \mid m, n \in \mathbb{N}\}$.
- $\mathbb{Q} := \left\{ \frac{m}{n} \mid m \in \mathbb{Z}, n \in \mathbb{N} \right\}$.

Also, we denote the set of positive numbers or negative numbers as follows:

- $\mathbb{Z}^+ := \{n \mid n \in \mathbb{N}\}$, $\mathbb{Z}^- := \{-n \mid n \in \mathbb{N}\}$.
- $\mathbb{Q}^+ := \left\{ \frac{m}{n} \mid m \in \mathbb{N}, n \in \mathbb{N} \right\}$, $\mathbb{Q}^- := \left\{ \frac{m}{n} \mid m \in \mathbb{Z}^-, n \in \mathbb{N} \right\}$.

1.2 Ordered Sets

A set S is called an **ordered set** if it is equipped with a relation $<$ such that for any $x, y, z \in S$, we have the following properties:

(A) $x < y$ and $y < z$ implies $x < z$.

(B) Only one of the relations $x < y$, $x = y$, or $x > y$ holds.

$x > y$ means $y < x$, $x \leq y$ means $(x < y \text{ or } x = y)$ and $x \geq y$ means $y \leq x$.

Remark. In an ordered set S , the relation $<$ is called an **order** (or a **total order**) on S . If a relation $<$ on T satisfies (A) only, then the relation $<$ is called a **partial order** on T and the set T is called **partially ordered**.

Definition 1.2.1. Let S be an ordered set and $E \subseteq S$.

- E is **bounded above** if there exists some $M \in S$ such that $x \leq M$ for all $x \in E$. Such an M is called an **upper bound** of E .
- E is **bounded below** if there exists some $m \in S$ such that $m \leq x$ for all $x \in E$. Such an m is called a **lower bound** of E .

Theorem 1.2.2. Any nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{Z}$ has a unique least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{Z}$ has a unique greatest element.

Proof. Let E be a nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{Z}$. Then there exists a lower bound $m \in \mathbb{Z}$ of E . Let F be defined as follows:

$$F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}.$$

By [Theorem 1.1.1](#), F has a least element n . Then we have

$$n \leq x - m + 1 \text{ for all } x \in E \implies n + m - 1 \leq x \text{ for all } x \in E.$$

Since $n \in F$, there exists a unique $x_0 \in E$ such that $n = x_0 - m + 1$, so $x_0 = n + m - 1$ is the unique least element of E .

Let S be a nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{Z}$. Then there exists an upper bound $M \in \mathbb{Z}$ of S . Let T be defined as follows:

$$T := \{-x + M + 1 \mid x \in S\} \subseteq \mathbb{N}.$$

By [Theorem 1.1.1](#), T has a least element n . Then we have

$$n \leq -x + M + 1 \text{ for all } x \in S \implies M - n + 1 \geq x \text{ for all } x \in S.$$

Since $n \in T$, there exists a unique $x_0 \in S$ such that $n = -x_0 + M + 1$, so $x_0 = M - n + 1$ is the unique greatest element of S . $\square$

Definition 1.2.3. Let S be an ordered set. Assume that $E \subseteq S$ is bounded above. If there exists an $\alpha \in S$ such that

- α is an upper bound of E .
- If $\beta < \alpha$, then β is not an upper bound of E .

then α is called the **supremum** (or *least upper bound*) of E , denoted by

$$\alpha = \sup E.$$

Similarly, Assume that $F \subseteq S$ is bounded below. If there exists an $\gamma \in F$ such that

- γ is a lower bound of F .
- If $\beta > \gamma$, then β is not a lower bound of F .

then γ is called the **infimum** (or *greatest lower bound*) of F , denoted by

$$\gamma = \inf F.$$

Example 1.2.4. Let $S \subseteq \mathbb{Q}^+$ be defined as

$$S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}.$$

For any $p \in S$, let

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p.$$

Then we have $q \in \mathbb{Q}^+$ and

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} < 0 \implies q \in S.$$

This implies that any $p \in S$ is not an upper bound of S .

For any $p \in \mathbb{Q}^+$ with $p^2 > 2$, p is an upper bound of S . Let

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p.$$

Then we have $q \in \mathbb{Q}^+$ and

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0 \implies q \text{ is a lower bound of } S.$$

This implies that such p is not the least upper bound of S . Since there exists no $p \in \mathbb{Q}^+$ such that $p^2 = 2$, S has no least upper bound in $\mathbb{Q}^+$.

Definition 1.2.5. An ordered set S is said to have the **least upper bound property** if any nonempty subset of S which is bounded above has a least upper bound in S .

Theorem 1.2.6. Let S be an ordered set with the least upper bound property. Then, any nonempty subset of S which is bounded below has a greatest lower bound in S .

Proof. Let E be a nonempty subset of S which is bounded below. Let L be defined as follows:

$$L := \{\lambda \mid \lambda \text{ is a lower bound of } E\}.$$

Then, L is nonempty and bounded above. By the least upper bound property, there exists $\alpha = \sup L \in S$. We will show that $\alpha = \inf E$.

Assume that $\alpha \notin L$. Then we have the following:

- α is not a lower bound of $E \implies$ There exists $\beta \in E$ such that $\beta < \alpha$.
- $\alpha = \sup L \implies \beta$ is not an upper bound of $L \implies$ There exists $\gamma \in L$ such that $\beta < \gamma \leq \alpha$.
- $\gamma \in L \implies \gamma$ is a lower bound of $E \implies \gamma \leq \beta$.

This is a contradiction. Therefore, we have $\alpha \in L$, i.e., α is a lower bound of E . Since α is an upper bound of L , the following holds:

$$\text{For any } \delta > \alpha, \text{ we have } \delta \notin L \implies \delta \text{ is not a lower bound of } E.$$

Therefore, we can conclude that $\alpha = \inf E$. □

1.3 Fields and Ordered Fields

A set F is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$ for all $x, y \in F$.
- $x + y = y + x$ for all $x, y \in F$.
- $(x + y) + z = x + (y + z)$ for all $x, y, z \in F$.
- There exists an element $0 \in F$ such that $x + 0 = x$ for all $x \in F$.
- For each $x \in F$, there exists an element $-x \in F$ such that $x + (-x) = 0$.

(B) Axioms for multiplication

- $x \cdot y \in F$ for all $x, y \in F$.

- $x \cdot y = y \cdot x$ for all $x, y \in F$.
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in F$.
- There exists an element $1 \in F$ with $1 \neq 0$ such that $x \cdot 1 = x$ for all $x \in F$.
- For each $x \in F$ with $x \neq 0$, there exists an element $x^{-1} \in F$ such that $x \cdot x^{-1} = 1$.

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in F$.

A set F is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If $x, y, z \in F$ and $x < y$, then $x + z < y + z$.
- If $x, y \in F$ and $x > 0$ and $y > 0$, then $x \cdot y > 0$.

Example 1.3.1. $(\mathbb{Q}, +, \cdot)$ is an ordered field, while $(\mathbb{Z}, +, \cdot)$ is not an ordered field since $2^{-1} \notin \mathbb{Z}$.

1.4 The Real Field

There exists an ordered field $(\mathbb{R}, +, \cdot)$ which has the [least upper bound property](#). Moreover, $\mathbb{R}$ contains $\mathbb{Q}$ as a subfield, i.e., $\mathbb{Q} \subseteq \mathbb{R}$ and the operations of addition and multiplication on $\mathbb{R}$ coincide with those on $\mathbb{Q}$.

We can construct $\mathbb{R}$ from $\mathbb{Q}$. The construction of $\mathbb{R}$ is rather complicated, so it will be discussed separately in [Appendix](#). In this section, we will just assume that $\mathbb{R}$ exists.

Theorem 1.4.1. *For any $x \in \mathbb{R}^+$ and $y \in \mathbb{R}$, there exists some $n \in \mathbb{N}$ such that $nx > y$. This theorem is called the **Archimedean property** of $\mathbb{R}$.*

Proof. Let $x \in \mathbb{R}^+$ and $y \in \mathbb{R}$ be given. Let E be defined as follows:

$$E := \{nx \mid nx \leq y, n \in \mathbb{N}\}.$$

If $E = \emptyset$, which implies that $x > y$, then $n = 1$ satisfies $nx > y$. Otherwise, E is nonempty and bounded above by y . By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}$. Then we have the following:

- $\alpha - x$ is not an upper bound of $E \implies$ There exists some $m \in \mathbb{N}$ such that $mx > \alpha - x$.

$$\bullet \quad mx > \alpha - x \implies (m+1)x > \alpha \implies (m+1)x \notin E \implies (m+1)x > y.$$

Therefore, we can take $n = m + 1$ to complete the proof. $\square$

Corollary 1.4.2. *For any $\varepsilon \in \mathbb{R}^+$ and $b \in (1, +\infty)$, there exists some $n \in \mathbb{N}$ such that $b^{-n} < \varepsilon$.*

Proof. For any $n \in \mathbb{N}$ and $x \in [-1, +\infty)$, we have

$$(1+x)^n \geq 1+nx.$$

Let $a \in \mathbb{R}^+$ and $\varepsilon \in \mathbb{R}^+$ be given. By [Theorem 1.4.1](#), there exists some $n \in \mathbb{N}$ such that $na > \frac{1}{\varepsilon}$. Then we have

$$(1+a)^n \geq 1+na > na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon.$$

Since $b > 1$, we can take $a = b - 1 > 0$ to complete the proof. $\square$

Theorem 1.4.3. *Any nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{R}$ has a unique least element, and any nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{R}$ has a unique greatest element.*

Proof. Let E be a nonempty subset of $\mathbb{Z}$ which is bounded below in $\mathbb{R}$. Then there exists a lower bound $\alpha \in \mathbb{R}$ of E . Then we have the following:

- If $\alpha \geq 0$, then $-1 \in \mathbb{Z}$ is a lower bound of E .
- If $\alpha < 0$, then by [Theorem 1.4.1](#), there exists some $n \in \mathbb{N}$ such that $n > -\alpha$, which implies that $-n \in \mathbb{Z}$ is a lower bound of E .

Therefore, E is bounded below in $\mathbb{Z}$. By [Theorem 1.2.2](#), E has a unique least element.

Let F be a nonempty subset of $\mathbb{Z}$ which is bounded above in $\mathbb{R}$. Then there exists an upper bound $\beta \in \mathbb{R}$ of F . Then we have the following:

- By [Theorem 1.4.1](#), there exists some $n \in \mathbb{N}$ such that $n > \beta$, which implies that $n \in \mathbb{Z}$ is an upper bound of F .

Therefore, F is bounded above in $\mathbb{Z}$. By [Theorem 1.2.2](#), F has a unique greatest element. $\square$

Definition 1.4.4. For any $x \in \mathbb{R}$, the **floor function** $\lfloor x \rfloor$ and **ceiling function** $\lceil x \rceil$ are defined as follows:

- $\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}.$
- $\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}.$

Theorem 1.4.3 guarantees the *existence* and *uniqueness* of $\lfloor x \rfloor$ and $\lceil x \rceil$.

Theorem 1.4.5. For any $-\infty < x < y < +\infty$, there exists some $p \in \mathbb{Q}$ such that $x < p < y$. This theorem states that $\mathbb{Q}$ is **dense** in $\mathbb{R}$.

Proof. Let $\varepsilon := y - x > 0$. By **Theorem 1.4.1**, there exists some $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$. Let S be defined as follows:

$$S := \left\{ k \in \mathbb{Z} \mid \frac{k}{n} < x \right\}.$$

S is nonempty since $n\lfloor x \rfloor - 1 \in S$, and is bounded above by $nx \in \mathbb{R}$. By **Theorem 1.4.3**, S has a greatest element, say m .

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take $p = \frac{m+1}{n} \in \mathbb{Q}$ to complete the proof. $\square$

Theorem 1.4.6. For each $x \in \mathbb{R}^+$ and $n \in \mathbb{N}$, there exists a unique $y \in \mathbb{R}^+$ such that $y^n = x$. We denote y by $\sqrt[n]{x}$ or $x^{\frac{1}{n}}$.

Proof. Let E be defined as follows:

$$E := \{t \in \mathbb{R}^+ \mid t^n < x\}.$$

E is nonempty since $2^{-1} \in E$ if $x \geq 1$, and $x^2 \in E$ otherwise. Also, E is bounded above by $\max\{x, 1\}$. By the least upper bound property, there exists $\alpha = \sup E \in \mathbb{R}^+$. We will show that $\alpha^n = x$.

For any $a, b \in \mathbb{R}^+$ with $a < b$, we have

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a) n b^{n-1}.$$

Assume that $\alpha^n < x$. By **Theorem 1.4.5**, there exists some $h \in \mathbb{R}^+$ such that

$$h < \min \left\{ 1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}} \right\}.$$

Put $a = \alpha$ and $b = \alpha + h$. Then we have

$$\begin{aligned} (\alpha + h)^n - \alpha^n &< hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha + h)^n &< \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

Therefore, we have $\alpha + h \in E$, which is a contradiction.

Assume that $\alpha^n > x$. By [Theorem 1.4.5](#), there exists some $h \in \mathbb{R}^+$ such that

$$h < \min \left\{ 1, \alpha, \frac{\alpha^n - x}{n(\alpha + 1)^{n-1}} \right\}.$$

Put $a = \alpha - h$ and $b = \alpha$. Then we have

$$\begin{aligned} \alpha^n - (\alpha - h)^n &< hn\alpha^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha - h)^n &> \alpha^n - hn(\alpha + 1)^{n-1} > \alpha^n - (x - \alpha^n) = x. \end{aligned}$$

Therefore, we have $\alpha - h$ is an upper bound of E , which is a contradiction.

Therefore, we have $\alpha^n = x$, which guarantees the *existence* of y . For any $\alpha, \beta \in \mathbb{R}^+$, we have $\alpha \neq \beta \implies \alpha^n \neq \beta^n$, which guarantees the *uniqueness* of y . $\square$

Definition 1.4.7. Let $y \in [0, 1)$ and $b \in \mathbb{N}$ be given with $b \geq 2$. A sequence $\{a_n\}_{n=1}^\infty$ is called the **greedy base b expansion** of y if for each $n \in \mathbb{N}$, we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

where $a_k \in S_b := \{0, 1, 2, \dots, b-1\}$ for all $k \in \mathbb{N}$. Especially, if $b = 10$, the sequence is called the **greedy decimal expansion** of y .

Theorem 1.4.8. For any $y \in [0, 1)$, the greedy base b expansion of y exists and is unique.

Proof. Let $\{t_n\}_{n=1}^\infty$, $\{T_n\}_{n=1}^\infty$ and $\{a_n\}_{n=1}^\infty$ be defined as follows:

- $t_1 := by$, $T_1 := \{t \in S_b \mid t \leq t_1\}$, $a_1 := \max T_1$.
 - $t_2 := b^2y - ba_1$, $T_2 := \{t \in S_b \mid t \leq t_2\}$, $a_2 := \max T_2$.
 - $\vdots$
 - $t_n := b^n y - \sum_{k=1}^{n-1} a_k b^{n-k}$, $T_n := \{t \in S_b \mid t \leq t_n\}$, $a_n := \max T_n$
- for all $n \in \mathbb{N}$ with $n \geq 2$,

Assume that $0 \leq t_n < b$ for some $n \in \mathbb{N}$. Then, since $0 \in T_n$ and T_n is bounded above, T_n has a greatest element by [Theorem 1.2.2](#), which implies that $\{a_n\}_{n=1}^\infty$ is well-defined. Let $P(n)$ be the statement that

$$P(n) : 0 \leq t_n < b \text{ and } a_n \text{ is well-defined.}$$

- **Base case:** $P(1)$ is true since $0 \leq t_1 = by < b$.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Then we have

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^n a_k b^{n+1-k} \\ &= b \left(b^n y - \sum_{k=1}^n a_k b^{n-k} \right) \\ &= b \left(b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n \right) \\ &= b(t_n - a_n). \end{aligned}$$

Since $a_n = \max T_n$ and $t_n < b$, we have

$$a_n \leq t_n < a_n + 1 \implies 0 \leq t_n - a_n < 1 \implies 0 \leq t_{n+1} = b(t_n - a_n) < b.$$

Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. By the definition of a_n , we have

$$\begin{aligned} a_n &\leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \implies 0 &\leq b^n y - \sum_{k=1}^n a_k b^{n-k} < 1 \\ \implies 0 &\leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \end{aligned}$$

for each $n \in \mathbb{N}$. Therefore, the *existence* of the *greedy base b expansion* of y is guaranteed.

Next, assume that there exist two sequences $\{a_n\}_{n=1}^\infty$ and $\{c_n\}_{n=1}^\infty$ such that for each $n \in \mathbb{N}$, we have

$$\bullet \quad 0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

$$\bullet \quad 0 \leq y - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n}$$

where $a_k \in S_b$ and $c_k \in S_b$ for all $k = 1, 2, \dots, n$. We will show that for each $n \in \mathbb{N}$, we have $a_n = c_n$. Let $Q(n)$ be the statement that

$$Q(n) : a_k = c_k \text{ for all } k = 1, 2, \dots, n.$$

• **Base case:** We have

$$\begin{aligned} - \quad 0 \leq y - \frac{a_1}{b} < \frac{1}{b} &\implies \frac{a_1}{b} \leq y < \frac{a_1 + 1}{b}. \\ - \quad 0 \leq y - \frac{c_1}{b} < \frac{1}{b} &\implies \frac{c_1}{b} \leq y < \frac{c_1 + 1}{b}. \end{aligned}$$

By combining the two inequalities, we have

$$\begin{aligned} - \quad \frac{a_1}{b} \leq y < \frac{c_1 + 1}{b} &\implies a_1 < c_1 + 1 \implies a_1 \leq c_1. \\ - \quad \frac{c_1}{b} \leq y < \frac{a_1 + 1}{b} &\implies c_1 < a_1 + 1 \implies c_1 \leq a_1. \end{aligned}$$

Therefore, we have $a_1 = c_1$, which shows that $Q(1)$ is true.

• **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. We have

$$\begin{aligned} - \quad 0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}}. \\ - \quad 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}}. \end{aligned}$$

By combining the two inequalities, we have

$$\begin{aligned} - \quad \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}}. \\ - \quad \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}}. \end{aligned}$$

By the inductive hypothesis, we have $a_k = c_k$ for all $k = 1, 2, \dots, n$. Therefore, we have

$$\begin{aligned} - \quad \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies a_{n+1} - c_{n+1} < 1 \implies a_{n+1} \leq c_{n+1}. \\ - \quad \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies c_{n+1} - a_{n+1} < 1 \implies c_{n+1} \leq a_{n+1}. \end{aligned}$$

Therefore, we have $a_{n+1} = c_{n+1}$, which shows that $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$. Therefore, the uniqueness of the greedy base b expansion of y is guaranteed. $\square$

Theorem 1.4.9. *Let $b \in \mathbb{N}$ be given with $b \geq 2$. Let $\{a_n\}_{n=1}^{\infty}$ be the greedy base b expansion of $a \in [0, 1)$, and let $\{c_n\}_{n=1}^{\infty}$ be the greedy base b expansion of $c \in [0, 1)$. If $a_n = c_n$ for all $n \in \mathbb{N}$, then we have $a = c$.*

Proof. Assume that $a \neq c$. Without loss of generality, assume that $a < c$. Then, by Corollary 1.4.2, there exists some $n \in \mathbb{N}$ such that

$$\frac{1}{b^n} < c - a.$$

By the definition of the greedy base b expansion, we have

$$\begin{aligned} \bullet \quad 0 \leq a - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} &\implies \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n}. \\ \bullet \quad 0 \leq c - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} &\implies c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}. \end{aligned}$$

By the assumption that $a_n = c_n$ for all $n \in \mathbb{N}$, we have

$$\sum_{k=1}^n \frac{a_k}{b^k} = \sum_{k=1}^n \frac{c_k}{b^k}.$$

Therefore, we have a contradiction:

$$\sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}.$$

Therefore, we have $a = c$, which completes the proof. $\square$

Theorem 1.4.10. *Let $b \in \mathbb{N}$ be given with $b \geq 2$, and let $\{a_n\}_{n=1}^{\infty}$ be given with $a_k \in S_b$ for all $k \in \mathbb{N}$. If there exists no $y \in [0, 1)$ such that*

$$\{a_n\}_{n=1}^{\infty} \text{ is the greedy base } b \text{ expansion of } y,$$

then there exists some $N \in \mathbb{N}$ such that

$$a_n = b - 1 \quad \text{for all } n \geq N.$$

Proof. We will prove the contrapositive of the statement: For all $N \in \mathbb{N}$, if there exists some $n \in \mathbb{N}$ such that

$$n \geq N \text{ and } a_n \neq b - 1,$$

then there exists $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the greedy base b expansion of y .

Assume that the condition of the above claim holds. Let $x_n := \sum_{k=1}^n \frac{a_k}{b^k}$ for each $n \in \mathbb{N}$. Since $a_k \in S_b$ for all $k \in \mathbb{N}$, we have

$$0 \leq x_n \leq \sum_{k=1}^n \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1 \quad \text{for all } n \in \mathbb{N}.$$

Then, we have that $X := \{x_n \mid n \in \mathbb{N}\}$ is bounded above. By the [least upper bound property](#), there exists $y = \sup X \geq 0$.

First, we will show that $y < 1$. By the assumption, there exists $m \in \mathbb{N}$ such that $a_m \neq b - 1$. Then, for all $n > m$, we have

$$\begin{aligned} x_n &= x_m + \sum_{k=m+1}^n \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^n \frac{b-1}{b^k} \\ \implies x_n &\leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m}. \end{aligned}$$

Since $a_m \leq b - 2$, we have

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^m \frac{b-1}{b^k} = 1 - \frac{1}{b^m}. \end{aligned}$$

We have $x_n \leq x_m$ for all $n \leq m$, and $x_n < x_m + \frac{1}{b^m} \leq 1 - \frac{1}{b^m}$ for all $n > m$.

Thus $1 - \frac{1}{b^m}$ is an upper bound of X . Since $y = \sup X$, we have

$$y \leq 1 - \frac{1}{b^m} < 1 \implies y \in [0, 1).$$

Next, we will show that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y . Let $n \in \mathbb{N}$ be fixed. By the assumption, there exists $m \in \mathbb{N}$ such that $m \geq n$ and $a_m \leq b - 2$.

$$\begin{aligned} x_m &= x_n + \sum_{k=n+1}^m \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^m \frac{b-1}{b^k} \\ \implies x_m &< x_n + \frac{1}{b^n} - \frac{1}{b^m} \implies x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}. \end{aligned}$$

Since $a_m \leq b - 2$, $x_m + \frac{1}{b^m}$ is an upper bound of X by the same argument as above. Since $y = \sup X$, we have

$$y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \implies 0 \leq y - x_n < \frac{1}{b^n}.$$

Therefore, $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y . $\square$

Theorem 1.4.11. Let $b \in \mathbb{N}$ be given with $b \geq 2$ and let $\{a_n\}_{n=1}^{\infty}$ be given with $a_k \in S_b$ for all $k \in \mathbb{N}$. If there exists some $N \in \mathbb{N}$ such that

$$a_n = b - 1 \quad \text{for all } n \geq N,$$

then there exists no $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y . This theorem is the converse of [Theorem 1.4.10](#).

Proof. Assume that there exists some $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y . Then we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \quad \text{for all } n \in \mathbb{N}.$$

Let $N \in \mathbb{N}$ be such that $a_n = b - 1$ for all $n \geq N$. Let $x_n := \sum_{k=1}^n \frac{a_k}{b^k}$ for each $n \in \mathbb{N}$. Then, for all $n \geq N$, we have

$$\begin{aligned} x_n &= x_{N-1} + \sum_{k=N}^n \frac{b-1}{b^k} = x_{N-1} + \frac{1}{b^{N-1}} - \frac{1}{b^n} \\ \implies y - x_n &= y - x_{N-1} - \frac{1}{b^{N-1}} + \frac{1}{b^n} \geq 0 \\ \implies \frac{1}{b^n} &\geq x_{N-1} + \frac{1}{b^{N-1}} - y. \end{aligned}$$

Since $0 \leq y - x_{N-1} < \frac{1}{b^{N-1}}$, we have $x_{N-1} + \frac{1}{b^{N-1}} - y > 0$. By [Corollary 1.4.2](#), there exists some $n \in \mathbb{N}$ with $n \geq N$ such that

$$\frac{1}{b^n} < x_{N-1} + \frac{1}{b^{N-1}} - y,$$

which is a contradiction. Therefore, there exists no $y \in [0, 1)$ such that $\{a_n\}_{n=1}^{\infty}$ is the *greedy base b expansion* of y . $\square$

1.5 The Extended Real Number System

The Extended real number system $\overline{\mathbb{R}}$ is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on $\overline{\mathbb{R}}$:

- $x + (+\infty) := +\infty$ for all $x \in \mathbb{R} \cup \{+\infty\}$.
- $x + (-\infty) := -\infty$ for all $x \in \mathbb{R} \cup \{-\infty\}$.
- $x \cdot (\pm\infty) := \pm\infty$ for all $x \in \mathbb{R}^+ \cup \{+\infty\}$.
- $x \cdot (\pm\infty) := \mp\infty$ for all $x \in \mathbb{R}^- \cup \{-\infty\}$.
- $\frac{x}{\pm\infty} := 0$ for all $x \in \mathbb{R}$.

The following operations are undefined in $\overline{\mathbb{R}}$:

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\frac{(\pm\infty)}{(\pm\infty)}$ and $\frac{(\pm\infty)}{(\mp\infty)}$
- $\frac{x}{0}$ for all $x \in \overline{\mathbb{R}}$

$\overline{\mathbb{R}}$ is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

Remark. We can identify the following intervals in $\overline{\mathbb{R}}$:

- $(-\infty, a) = \{x \in \mathbb{R} \mid x < a\}$ for all $a \in \mathbb{R}$.
- $(a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$ for all $a \in \mathbb{R}$.
- $(-\infty, +\infty) = \mathbb{R}$.

Theorem 1.5.1. *Any subset of $\overline{\mathbb{R}}$ has a least upper bound and a greatest lower bound in $\overline{\mathbb{R}}$.*

Proof. Let $E \subseteq \overline{\mathbb{R}}$. If $E = \emptyset$, then we have

- $\sup E = -\infty$

- $\inf E = +\infty$.

Otherwise, we have the following cases:

- If E is bounded above in $\mathbb{R}$, then $\sup E$ exists in $\mathbb{R}$ by the [least upper bound property](#) of $\mathbb{R}$.
- If E is not bounded above in $\mathbb{R}$, then $\sup E = +\infty$.
- If E is bounded below in $\mathbb{R}$, then $\inf E$ exists in $\mathbb{R}$ by [Theorem 1.2.6](#).
- If E is not bounded below in $\mathbb{R}$, then $\inf E = -\infty$.

□

1.6 The Complex Field

The complex field $\mathbb{C}$ is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on $\mathbb{C}$:

- $(x, y) + (u, v) := (x + u, y + v)$ for all $(x, y), (u, v) \in \mathbb{C}$.
- $(x, y) \cdot (u, v) := (xu - yv, xv + yu)$ for all $(x, y), (u, v) \in \mathbb{C}$.

$\mathbb{C}$ is a field with the following properties:

- The additive identity is $(0, 0)$ and the multiplicative identity is $(1, 0)$.
- The additive inverse of (x, y) is $(-x, -y)$ and the multiplicative inverse of (x, y) is

$$\left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

for all $(x, y) \in \mathbb{C}$ with $(x, y) \neq (0, 0)$.

Remark. The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \rightarrow \mathbb{R}$$

defined by $f((x, 0)) = x$ is an *isomorphism*. We can identify $\mathbb{R}$ with the subfield $\{(x, 0) \mid x \in \mathbb{R}\}$ of $\mathbb{C}$. We denote $(x, 0) \in \mathbb{C}$ by $x \in \mathbb{R}$.

Remark. We define $i := (0, 1) \in \mathbb{C}$. Then we have $i^2 = -1$, and any complex number $(x, y) \in \mathbb{C}$ can be written as

$$(x, y) = x + yi.$$

Definition 1.6.1. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. The **real part** of z is defined as

$$\Re(z) := x \text{ or } \Re(z) := x,$$

the **imaginary part** of z is defined as

$$\Im(z) := y \text{ or } \Im(z) := y,$$

and the **complex conjugate** of z is defined as

$$\bar{z} := x - yi.$$

Theorem 1.6.2. Let $z = x + yi \in \mathbb{C}$ and let $w = u + vi \in \mathbb{C}$ for some $x, y, u, v \in \mathbb{R}$. Then we have the following properties:

- $\overline{(\bar{z})} = z.$
- $z = \bar{z} \iff z \in \mathbb{R}.$
- $\overline{z + w} = \bar{z} + \bar{w}.$
- $\overline{zw} = \bar{z} \cdot \bar{w}.$
- $z + \bar{z} = 2\Re(z)$ and $z - \bar{z} = 2i\Im(z).$
- $z\bar{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}.$

Proof. The proof follows directly from the definitions and properties of the operations on $\mathbb{C}$. $\square$

Definition 1.6.3. Let $z = x + yi \in \mathbb{C}$ for some $x, y \in \mathbb{R}$. the **modulus** of z is defined as

$$|z| := \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The existence and uniqueness of $|z|$ follows from [Theorems 1.4.6](#) and [1.6.2](#).

Theorem 1.6.4. Let $a_1, a_2, \dots, a_n \in \mathbb{C}$ and $b_1, b_2, \dots, b_n \in \mathbb{C}$. Then we have

$$\left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2.$$

This theorem is called the **Cauchy-Schwarz inequality**.

Proof. Let $i, j \in \mathbb{N}$ with $1 \leq i < j \leq n$. Then we have

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned} \sum_{i < j} (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j) &= \sum_{i \neq j} (a_i \bar{b}_i \cdot \bar{a}_j b_j) \\ &= \sum_{i=1}^n a_i \bar{b}_i \cdot \sum_{j=1}^n \bar{a}_j b_j - \sum_{k=1}^n a_k \bar{b}_k \cdot \bar{a}_k b_k \\ &= \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

Therefore, we have

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \geq 0.$$

□

Corollary 1.6.5. Let $a_1, a_2, \dots, a_n \in \mathbb{R}$ and $b_1, b_2, \dots, b_n \in \mathbb{R}$. Then we have

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2.$$

Theorem 1.6.6. Let $z, w \in \mathbb{C}$. Then we have the following properties:

- (A) $|zw| = |z| \cdot |w|$.
- (B) $|z + w| \leq |z| + |w|$.
- (C) $|z| - |w| \leq |z - w|$.

Proof. Let $z = x + yi$ and $w = u + vi$ for some $x, y, u, v \in \mathbb{R}$.

(A) $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2.$

(B) By Corollary 1.6.5, we have $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2).$

$$\begin{aligned} |z + w|^2 &= (x + u)^2 + (y + v)^2 \\ &= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\ &\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| + |w|)^2. \end{aligned}$$

(C) Again, by Corollary 1.6.5, we have

$$\begin{aligned} |z - w|^2 &= (x - u)^2 + (y - v)^2 \\ &= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\ &\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\ &= (|z| - |w|)^2. \end{aligned}$$

□

1.7 Euclidean Spaces

For each $n \in \mathbb{N}$, we call the set of all ordered n -tuples of real numbers as the **Euclidean n -space**, and denote it as $\mathbb{R}^n$. We can identify $\mathbb{R}^n$ as

$$\mathbb{R}^n = \{\mathbf{x} = (x_1, x_2, \dots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \dots, n\},$$

where $x_1, x_2, \dots, x_n$ are called the **coordinates** of $\mathbf{x}$. The elements of $\mathbb{R}^n$ with $n > 1$ are called **points** (or **vectors**), which are denoted by boldface letters.

Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$, and let $\alpha \in \mathbb{R}$. We define the following operations on $\mathbb{R}^n$:

- $\mathbf{x} + \mathbf{y} := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n).$
- $\alpha \mathbf{x} := (\alpha x_1, \alpha x_2, \dots, \alpha x_n).$

Then, $\mathbb{R}^n$ is a *vector space* over $\mathbb{R}$ with the zero vector $\mathbf{0} = (0, 0, \dots, 0).$

Definition 1.7.1. Let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$. The **inner product** of $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$\mathbf{x} \cdot \mathbf{y} := \sum_{k=1}^n x_k y_k.$$

The **norm** of $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| := \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}.$$

The **distance** between $\mathbf{x}$ and $\mathbf{y}$ is defined as

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}.$$

We can confirm that $d(x, y) = |x - y|$ in the case of $n = 1$.

Theorem 1.7.2. *Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ for some $n \in \mathbb{N}$. Then we have the following properties:*

- (A) $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|.$
- (B) $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$
- (C) $\|\mathbf{x} - \mathbf{y}\| \geq \left| \|\mathbf{x}\| - \|\mathbf{y}\| \right|.$

Proof. (A) By [Corollary 1.6.5](#), we have

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2.$$

(B) By part (A), we have

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

(C) By part (B), we have

$$\begin{aligned} \|\mathbf{x}\| &= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|. \\ \|\mathbf{y}\| &= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| \implies \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

□

Chapter 2

Basic Topology

2.1 Relations

A **relation** $\sim$ on a set S defines a subset R of $S \times S$, and vice versa.

- $R := \{(x, y) \in S \times S \mid x \sim y\}$.
- $x \sim y \iff (x, y) \in R$.

Definition 2.1.1. A relation $\sim$ on a set S is called an **equivalence relation** if it satisfies the following properties:

- Reflexivity: $x \sim x$ for all $x \in S$.
- Symmetry: If $x \sim y$, then $y \sim x$ for all $x, y \in S$.
- Transitivity: If $x \sim y$ and $y \sim z$, then $x \sim z$ for all $x, y, z \in S$.

Definition 2.1.2. A **partition** of a set S is a collection $\{A_\alpha\}_{\alpha \in I}$ of nonempty subsets of S such that

- $A_\alpha \cap A_\beta = \emptyset$ for all $\alpha, \beta \in I$ with $\alpha \neq \beta$.
- $\bigcup_{\alpha \in I} A_\alpha = S$.

Definition 2.1.3. Let $\sim$ be an equivalence relation on a set S . For each $x \in S$, the **equivalence class** of x is defined as

$$[x] := \{y \in S \mid y \sim x\}.$$

We denote the set of all equivalence classes of S by $S/\sim := \{[x] \mid x \in S\}$.

Theorem 2.1.4. *Let $\sim$ be an equivalence relation on a set S . For any $x, y \in S$, we have either*

$$[x] = [y] \quad \text{or} \quad [x] \cap [y] = \emptyset.$$

Therefore, $S/\sim$ is a partition of S .

Proof. Assume that $[x] \cap [y] \neq \emptyset$. There exists some $z \in S$ such that $z \in [x]$ and $z \in [y]$. Then we have

$$z \sim x \text{ and } z \sim y \implies x \sim z \text{ and } z \sim y \implies x \sim y$$

by the symmetry and transitivity of $\sim$. Therefore, for any $w \in [x]$, we have

$$w \sim x \text{ and } x \sim y, \implies w \sim y \implies w \in [y].$$

This shows that $[x] \subseteq [y]$. Similarly, we can show that $[y] \subseteq [x]$. Therefore, we have $[x] = [y]$ whenever $[x] \cap [y] \neq \emptyset$. $\square$

Theorem 2.1.5. *Let $\{A_\alpha\}_{\alpha \in I}$ be a partition of a set S . Let $\sim$ be a relation on S defined by*

$$x \sim y \iff x, y \in A_\alpha \text{ for some } \alpha \in I.$$

Then, we have that $\sim$ is an equivalence relation on S , and the equivalence classes of $\sim$ are precisely the sets A_α .

Proof. We need to show that $\sim$ is reflexive, symmetric, and transitive.

- Reflexivity: For any $x \in S$, there exists $\alpha \in I$ such that $x \in A_\alpha$.
- Symmetry: If $x \sim y$, then there exists $\alpha \in I$ such that $y, x \in A_\alpha$.
- Transitivity: If $x \sim y$ and $y \sim z$, then there exist $\alpha, \beta \in I$ such that

$$x, y \in A_\alpha \text{ and } y, z \in A_\beta.$$

Since $y \in A_\alpha \cap A_\beta$ and the sets $\{A_\alpha\}_{\alpha \in I}$ are disjoint, we have $\alpha = \beta$, which implies that $x, z \in A_\alpha$.

Therefore, $\sim$ is an equivalence relation on S . The equivalence class of any $x \in S$ is precisely the set A_α containing x . $\square$

2.2 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain** X .
- A **codomain** Y .

- A **rule** f which associates each element $x \in X$ to a *unique* element $y \in Y$.

f is said to be a *function* of X into Y , which is denoted by $f : X \rightarrow Y$. For each $x \in X$, the *unique* element $y \in Y$ associated with x is denoted by $f(x)$.

Remark. Let $f : X \rightarrow Y$ be a function. For any subset $E \subseteq X$, we define the **image** of E under f as

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if $E = X$, then $f(X) \subseteq Y$ is called the **range** of f .

Remark. Let $f : X \rightarrow Y$ be a function. For any subset $E \subseteq Y$, we define the **inverse image** of E under f as

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X.$$

Definition 2.2.1. Let $f : X \rightarrow Y$ be a function. We define the properties of f as follows:

- f is **injective** (or *one-to-one*) if for any $x_1, x_2 \in X$, we have

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

- f is **surjective** (or *onto*) if $f(X) = Y$.
- f is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

Definition 2.2.2. Let $f : X \rightarrow Y$ be a bijective function. Then, for any $y \in Y$, there exists a *unique* $x \in X$ such that $f(x) = y$. Therefore, we can define a function

$$f^{-1} : Y \rightarrow X \quad \text{such that} \quad f^{-1}(y) = x.$$

Such a function f^{-1} is called the **inverse function** of f .

Theorem 2.2.3. Let $f : X \rightarrow Y$ be a bijective function. Then the inverse function $f^{-1} : Y \rightarrow X$ is bijective.

Proof. Let $y_1, y_2 \in Y$ be given. Assume that $f^{-1}(y_1) = f^{-1}(y_2) = x$. Then, by [Definition 2.2.2](#), we have

$$f(x) = y_1 \text{ and } f(x) = y_2 \implies y_1 = y_2$$

since f is a function. Therefore, we have that f^{-1} is injective. By the definition of image, we have that $f^{-1}(Y) = X$, which implies that f^{-1} is surjective. $\square$

Definition 2.2.4. Let $f : X \rightarrow Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ be given with $f(E) \subseteq F$. We define the restriction functions as follows:

- The **restriction** of f to E is defined by

$$f|_E : E \rightarrow Y \quad \text{such that} \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$

- The **Corestriction** of f to F is defined by

$$f|_F : X \rightarrow F \quad \text{such that} \quad f|_F(x) = f(x) \quad \text{for all } x \in X.$$

- The **Bi-restriction** of f to E and F is defined by

$$f|_E^F : E \rightarrow F \quad \text{such that} \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

Theorem 2.2.5. Let $f : X \rightarrow Y$ be a function. Let $E \subseteq X$ and $F \subseteq Y$ be given with $f(E) \subseteq F$. Then we have the following properties:

- (A) If f is injective, then $f|_E^F$ is also injective.
- (B) If $f(E) = F$, then $f|_E^F$ is surjective.
- (C) If f is injective and $f(E) = F$, then $f|_E^F$ is bijective.

Proof. (C) follows from (A) and (B). Let $x_1, x_2 \in E$.

- (A) Assume that f is injective. Then we have

$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

- (B) Assume that $f(E) = F$. Then, for any $y \in F$, there exists $x \in E$ such that $f(x) = y$. Therefore, we have $f|_E^F(E) = F$.

□

Definition 2.2.6. Let X, Y, Z be sets and $f : X \rightarrow Y, g : Y \rightarrow Z$ be functions. We define the **composition** of f and g as the function $g \circ f : X \rightarrow Z$ defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

Theorem 2.2.7. Let X, Y, Z be sets and $f : X \rightarrow Y, g : Y \rightarrow Z$ be functions. Then we have the following properties:

(A) If f and g are injective, then $g \circ f$ is also injective.

(B) If f and g are surjective, then $g \circ f$ is also surjective.

(C) If f and g are bijective, then $g \circ f$ is also bijective.

Proof. (C) follows from (A) and (B). Let $x_1, x_2 \in X$ be given.

(A) Assume that f and g are injective. Then we have

$$\begin{aligned} (g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ &\implies f(x_1) = f(x_2) \implies x_1 = x_2. \end{aligned}$$

(B) Assume that f and g are surjective. Then, for any $z \in Z$, there exists $y \in Y$ such that $g(y) = z$. Since f is surjective, there exists $x \in X$ such that $f(x) = y$. Therefore, we have $(g \circ f)(x) = g(f(x)) = g(y) = z$.

□

Theorem 2.2.8. Let S be a set of sets. For [Sections 2.3](#) and [2.4](#), we consider the relation $\sim$ defined on S by

$$X \sim Y \iff \text{There exists a bijection } f : X \rightarrow Y.$$

The relation $\sim$ is an equivalence relation on S .

Proof. Let $X, Y, Z \in S$ be given.

- **Reflexivity:** The identity function $id_X : X \rightarrow X$, defined by $id_X(x) = x$ for all $x \in X$, is bijective. Therefore, we have $X \sim X$.
- **Symmetry:** Assume that $X \sim Y$. Then there exists a bijection $f : X \rightarrow Y$. By [Theorem 2.2.3](#), the inverse function $f^{-1} : Y \rightarrow X$ is also bijective. Therefore, we have $Y \sim X$.
- **Transitivity:** Assume that $X \sim Y$ and $Y \sim Z$. Then there exist bijections $f : X \rightarrow Y$ and $g : Y \rightarrow Z$. By [Theorem 2.2.7](#), the composition $g \circ f : X \rightarrow Z$ is also bijective. Therefore, we have $X \sim Z$.

□

2.3 Finite Sets

Let J_n be a set defined by

$$J_n := \{1, 2, \dots, n\} \quad \text{for each } n \in \mathbb{N}.$$

A set X is said to be **finite** if $J_n \sim X$ for some $n \in \mathbb{N}$. In this case, n is called the **cardinality** of X , denoted by $|X|$. Especially, if $X = \emptyset$, we define $|X| = 0$. X is said to be **infinite** if X is not finite.

Lemma 2.3.1. *Any subset of J_n is finite for each $n \in \mathbb{N}$.*

Proof. Let $P(n)$ be the statement that any subset of J_n is finite.

- **Base case:** $J_1 = \{1\}$ has $\emptyset$ and J_1 as its only subsets. We have $J_1 \sim J_1$ by $f \equiv id_{J_1}$, and $\emptyset$ is finite by definition. Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let $A \subseteq J_{n+1}$ be given. We have two cases:
 - If $n+1 \notin A$, then $A \subseteq J_n$, which is finite by the inductive hypothesis.
 - If $n+1 \in A$, then $A = B \cup \{n+1\}$ for some $B \subseteq J_n$. By the inductive hypothesis, B is finite, so there exists a bijection $f : J_m \rightarrow B$ for some $m \in \mathbb{N}$. Let $g : J_{m+1} \rightarrow A$ be defined by

$$g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n+1 & \text{if } k = m+1 \end{cases}$$

Then g is bijective, which implies that A is finite.

Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.3.2. *Any subset of a finite set is finite.*

Proof. Let X be a finite set and $Y \subseteq X$. Then there exists a bijection $f : J_n \rightarrow X$ for some $n \in \mathbb{N}$. Let Z be defined by

$$Z := f^{-1}(Y) \subseteq J_n.$$

Since $Z \subseteq J_n$, Z is finite by [Lemma 2.3.1](#), so there exists a bijection $g : J_m \rightarrow Z$ for some $m \in \mathbb{N}$.

By [Theorem 2.2.5](#), $f|_Z^Y$ is bijective. Then, by [Theorem 2.2.7](#), the composition $f|_Z^Y \circ g : J_m \rightarrow Y$ is also bijective. Therefore, Y is finite. □

Lemma 2.3.3. *Let J_n be given for some $n \in \mathbb{N}$. If there exists an surjection $f : J_n \rightarrow Y$, then Y is finite.*

Proof. Let $f : J_n \rightarrow Y$ be a surjective function. For any $y \in Y$, let N_y be a set defined by

$$N_y := f^{-1}(\{y\}) \subseteq J_n.$$

Since f is surjective, N_y is nonempty. By [Theorem 1.1.1](#), N_y has a least element, say n_y . Let M_Y be a set defined by

$$M_Y := \{n_y \mid y \in Y\} \subseteq J_n.$$

Let $g : M_Y \rightarrow Y$ be a function defined by

$$g(n_y) = y \quad \text{for each } n_y \in M_Y.$$

Then g is bijective. By [Lemma 2.3.1](#), M_Y is finite, so there exists a bijection $h : J_m \rightarrow M_Y$ for some $m \in \mathbb{N}$. Then, by [Theorem 2.2.7](#), the composition $g \circ h : J_m \rightarrow Y$ is bijective. Therefore, Y is finite. $\square$

Theorem 2.3.4. *Let X be a finite set. If there exists an surjection $f : X \rightarrow Y$, then Y is finite.*

Proof. Let $f : X \rightarrow Y$ be a surjective function. Since X is finite, there exists a bijection $g : J_n \rightarrow X$ for some $n \in \mathbb{N}$. Then, by [Theorem 2.2.7](#), the composition $f \circ g : J_n \rightarrow Y$ is surjective. Therefore, by [Lemma 2.3.3](#), Y is finite. $\square$

Theorem 2.3.5. *Let X be a finite ordered set. Then X has both a least element and a greatest element.*

Proof. Since X is finite, there exists a bijection $f : J_n \rightarrow X$ for some $n \in \mathbb{N}$. The elements of X can be listed as $X = \{a_1, a_2, \dots, a_n\}$, where $a_k = f(k)$ for each $k = 1, 2, \dots, n$.

First, we will show that X has a least element. Let $P(n)$ be the statement that any finite ordered set of cardinality n has a least element.

- **Base case:** $X = \{a_1\}$ has a least element a_1 . Thus $P(1)$ is true.
- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$. Let X be a finite ordered set of cardinality $n + 1$ such that

$$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset $Y \subseteq X$ of cardinality n such that

$$Y = \{a_1, a_2, \dots, a_n\}$$

has a least element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:

- If $a_k \leq a_{n+1}$, then a_k is the least element of X .
- If $a_{n+1} < a_k$, then a_{n+1} is the least element of X .

Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$.

Next, we will show that X has a greatest element. Let $Q(n)$ be the statement that any finite ordered set of cardinality n has a greatest element.

- **Base case:** $X = \{a_1\}$ has a greatest element a_1 . Thus $Q(1)$ is true.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. Let X be a finite ordered set of cardinality $n+1$ such that

$$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset $Y \subseteq X$ of cardinality n such that

$$Y = \{a_1, a_2, \dots, a_n\}$$

has a greatest element, say a_k for some $k = 1, 2, \dots, n$. We have two cases:

- If $a_k \geq a_{n+1}$, then a_k is the greatest element of X .
- If $a_{n+1} > a_k$, then a_{n+1} is the greatest element of X .

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.3.6. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

- (A) *If there exists an injection $f : J_m \rightarrow J_n$, then we have $m \leq n$.*
- (B) *If there exists a surjection $f : J_m \rightarrow J_n$, then we have $m \geq n$.*
- (C) *If there exists a bijection $f : J_m \rightarrow J_n$, then we have $m = n$.*

This lemma guarantees the uniqueness of the cardinality. If $J_m \sim X$ and $J_n \sim X$, then we have $J_m \sim J_n$, which implies that $m = n$ by (C).

Proof. We will prove (A) by mathematical induction on m , (B) by mathematical induction on n , and (C) follows from (A) and (B).

- (A) Let $n \in \mathbb{N}$ be fixed. Let $P(m)$ be the statement that if there exists an injective function $f_m : J_m \rightarrow J_n$, then we have $m \leq n$.

- **Base case:** Let $m = 1$. Then $m \leq n$ holds. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(m)$ is true for some $m \in \mathbb{N}$. If $n = 1$, there does not exist an injective function $f_{m+1} : J_{m+1} \rightarrow J_n$ since $f_{m+1}(1) \neq f_{m+1}(m+1)$, where $P(m+1)$ is true.

Otherwise, assume that such a function f_{m+1} exists. We have two cases:

- If $f_{m+1}(m+1) = n$, let f_m be a function defined by

$$f_m \equiv f_{m+1}|_{J_m^{n-1}} : J_m \rightarrow J_{n-1}.$$

Then f_m is injective by [Theorem 2.2.5](#). By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

- If $f_{m+1}(m+1) = x \neq n$, let $g_n : J_n \rightarrow J_n$ be defined by

$$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then g_n is bijective. Let f_m be a function defined by

$$f_m \equiv (g_n \circ f_{m+1})|_{J_m^{n-1}} : J_m \rightarrow J_{n-1}.$$

Then f_m is injective by [Theorems 2.2.5](#) and [2.2.7](#). By the inductive hypothesis, we have $m \leq n-1$, which implies that $m+1 \leq n$.

Therefore, $P(m+1)$ is true whenever $P(m)$ is true.

By the principle of mathematical induction, $P(m)$ is true for all $m \in \mathbb{N}$.

- (B) Let $m \in \mathbb{N}$ be fixed. Let $Q(n)$ be the statement that if there exists a surjective function $f_n : J_m \rightarrow J_n$, then we have $m \geq n$.

- **Base case:** Let $n = 1$. Then $m \geq n$ holds. Thus $Q(1)$ is true.
- **Inductive step:** Assume that $Q(n)$ is true for some $n \in \mathbb{N}$. If $m = 1$, there does not exist a surjective function $f_{n+1} : J_m \rightarrow J_{n+1}$ since $f_{n+1}(1)$ is unique, where $Q(n+1)$ is true.

Otherwise, assume that such a function f_{n+1} exists. We have two cases:

- If $f_{n+1}(m) = n+1$, let K_{n+1} be a set defined by

$$K_{n+1} := (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let $f_n : J_m \rightarrow J_n$ be a function defined by

$$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then f_n is surjective. By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

- If $f_{n+1}(m) = x \neq n+1$, let $g_{n+1} : J_{n+1} \rightarrow J_{n+1}$ be a function defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then g_{n+1} is bijective. Let K_{n+1} be a set defined by

$$K_{n+1} := (g_{n+1} \circ f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let $f_n : J_m \rightarrow J_n$ be a function defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then f_n is surjective by [Theorem 2.2.7](#). By the inductive hypothesis, we have $m-1 \geq n$, which implies that $m \geq n+1$.

Thus $Q(n+1)$ is true whenever $Q(n)$ is true.

By the principle of mathematical induction, $Q(n)$ is true for all $n \in \mathbb{N}$.

□

Lemma 2.3.7. *Let J_m, J_n be given for some $m, n \in \mathbb{N}$. Then we have the following properties:*

- (A) *If $m \leq n$, then there exists an injection $f : J_m \rightarrow J_n$.*
- (B) *If $m \geq n$, then there exists a surjection $f : J_m \rightarrow J_n$.*
- (C) *If $m = n$, then there exists a bijection $f : J_m \rightarrow J_n$.*

This lemma is the converse of [Lemma 2.3.6](#).

Proof. We will prove this lemma by explicitly constructing the functions.

- (A) If $m \leq n$, let $f : J_m \rightarrow J_n$ be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then f is an injection.

- (B) If $m \geq n$, let $f : J_m \rightarrow J_n$ be a function defined by

$$f(i) = \begin{cases} i, & \text{if } i \leq n \\ n, & \text{if } i > n \end{cases}$$

Then f is a surjection.

(C) If $m = n$, let $f : J_m \rightarrow J_n$ be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then f is a bijection.

□

Theorem 2.3.8. *Let X, Y be finite sets. Then we have the following properties:*

(A) *If there exists an injection $f : X \rightarrow Y$, then we have $|X| \leq |Y|$.*

(B) *If there exists a surjection $f : X \rightarrow Y$, then we have $|X| \geq |Y|$.*

(C) *If there exists a bijection $f : X \rightarrow Y$, then we have $|X| = |Y|$.*

Moreover, the converses of (A), (B), and (C) are also true.

*The contraposition of (A) is called the **pigeonhole principle**, which states that if $|X| > |Y|$, then there exists no injection $f : X \rightarrow Y$.*

Proof. Let $|X| = m$ and $|Y| = n$ for some $m, n \in \mathbb{N}$. Then there exist bijections $f_X : J_m \rightarrow X$ and $f_Y : J_n \rightarrow Y$. By Theorems 2.2.3 and 2.2.7, the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \rightarrow J_n$$

is injective, surjective, or bijective if f is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.6, we have the desired results:

(A) If f is injective, then F is injective, which implies that $m \leq n$.

(B) If f is surjective, then F is surjective, which implies that $m \geq n$.

(C) If f is bijective, then F is bijective, which implies that $m = n$.

Also, by Theorems 2.2.3 and 2.2.7, the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \rightarrow Y$$

is injective, surjective, or bijective if $G : J_m \rightarrow J_n$ is injective, surjective, or bijective, respectively. Then, by Lemma 2.3.7, we have the desired converses:

(A) If $m \leq n$, then there exists an injective function $G : J_m \rightarrow J_n$, which implies that there exists an injective function $f : X \rightarrow Y$.

(B) If $m \geq n$, then there exists a surjective function $G : J_m \rightarrow J_n$, which implies that there exists a surjective function $f : X \rightarrow Y$.

(C) If $m = n$, then there exists a bijective function $G : J_m \rightarrow J_n$, which implies that there exists a bijective function $f : X \rightarrow Y$.

□

Lemma 2.3.9. *Let X, Y be finite sets with $X \cap Y = \emptyset$. Then the union $X \cup Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \rightarrow X$ and $g : J_n \rightarrow Y$. Let $h : J_{m+n} \rightarrow X \cup Y$ be a function defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i - m), & \text{if } m < i \leq m + n \end{cases}$$

Then h is bijective, which implies that $X \cup Y$ is finite.

□

Theorem 2.3.10. *Let X, Y be finite sets. Then the union $X \cup Y$ is finite.*

Proof. Let A, B, C be sets defined by

- $A = X \setminus Y \subseteq X$.
- $B = Y \setminus X \subseteq Y$.
- $C = X \cap Y \subseteq X$.

Then A, B, C are finite by [Theorem 2.3.2](#) and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$.
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$.
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$.

By [Lemma 2.3.9](#), $X \cup Y$ is finite.

□

Theorem 2.3.11. *Let $X_1, X_2, \dots, X_n$ be finite sets for some $n \in \mathbb{N}$. Then the union $\bigcup_{k=1}^n X_k$ is finite, i.e., any finite union of finite sets is finite.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets} \implies \bigcup_{k=1}^n X_k \text{ is finite.}$$

- **Base case:** Let X_1 be a finite set. Then $\bigcup_{k=1}^1 X_k = X_1$ is finite. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be finite sets. Then we have

$$\bigcup_{k=1}^{n+1} X_k = \left(\bigcup_{k=1}^n X_k \right) \cup X_{n+1}.$$

By the inductive hypothesis, $\bigcup_{k=1}^n X_k$ is finite. Then, by [Theorem 2.3.10](#),

$\bigcup_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Lemma 2.3.12. *Let X, Y be finite sets. Then the cartesian product $X \times Y$ is finite.*

Proof. Let $m = |X|$ and $n = |Y|$. There exist bijections $f : J_m \rightarrow X$ and $g : J_n \rightarrow Y$. Let $h : J_{mn} \rightarrow X \times Y$ be a function defined by

$$h(i) = \left(f\left(\left[\frac{i}{n}\right]\right), g\left(i - n\left(\left[\frac{i}{n}\right] - 1\right)\right) \right) \text{ for all } i \in J_{mn}.$$

Then h is bijective, which implies that $X \times Y$ is finite. □

Theorem 2.3.13. *Let $X_1, X_2, \dots, X_n$ be finite sets for some $n \in \mathbb{N}$. Then the cartesian product $\prod_{k=1}^n X_k$ is finite.*

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets} \implies \prod_{k=1}^n X_k \text{ is finite.}$$

- **Base case:** Let X_1 be a finite set. Then, $\prod_{k=1}^1 X_k = X_1$ is finite. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be finite sets. Then we have

$$\prod_{k=1}^{n+1} X_k = \left(\prod_{k=1}^n X_k \right) \times X_{n+1}.$$

By the inductive hypothesis, $\prod_{k=1}^n X_k$ is finite. Then, by [Lemma 2.3.12](#), $\prod_{k=1}^{n+1} X_k$ is finite. Thus $P(n+1)$ is true whenever $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. $\square$

Theorem 2.3.14. *Let X be a nonempty set. Then there exists no surjection $f : X \rightarrow \mathcal{P}(X)$, where*

$$\mathcal{P}(X) := \{E \mid E \subseteq X\}$$

*is called the **power set** of X . This theorem is called **Cantor's theorem**.*

Proof. Let S be a set defined by

$$S := \{x \in X \mid x \notin f(x)\}.$$

Since $S \subseteq X$, we have that $S \in \mathcal{P}(X)$. Assume that there exists a surjection $f : X \rightarrow \mathcal{P}(X)$. Then there exists $x \in X$ such that $f(x) = S$.

If $x \in S$, then we have $x \notin f(x) = S$, which is a contradiction. If $x \notin S$, then we have $x \in f(x) = S$, which is also a contradiction. Therefore, there exists no surjection $f : X \rightarrow \mathcal{P}(X)$. $\square$

2.4 Countable and Uncountable Sets

A set X is said to be **countable** if $\mathbb{N} \sim X$. A set X is said to be **uncountable** if it is neither finite nor countable. A set X is said to be **at most countable** if X is either finite or countable.

Theorem 2.4.1. *Let X be an infinite set, and let Y be a finite set. Then $X \setminus Y$ is infinite.*

Proof. Assume that $X \setminus Y$ is finite. Since Y is finite, by [Theorem 2.3.10](#), $(X \setminus Y) \cup Y$ is finite. Then, by [Theorem 2.3.2](#), $X \subseteq (X \setminus Y) \cup Y$ is finite, which contradicts the assumption. Therefore, $X \setminus Y$ is infinite. $\square$

Theorem 2.4.2. *Let X be an infinite set. If there exists a injection $f : X \rightarrow Y$, then Y is infinite.*

Proof. Assume that Y is finite. Let $g : Y \rightarrow X$ be defined by

$$g(y) = \begin{cases} f^{-1}(y), & \text{if } y \in f(X) \\ f^{-1}(y_0), & \text{Otherwise} \end{cases}$$

where $y_0 \in f(X)$ is fixed. Then g is surjective. By [Theorem 2.3.4](#), X is finite, which is a contradiction. Therefore, Y is infinite. $\square$

Lemma 2.4.3. *$\mathbb{N}$ is infinite.*

Proof. Assume that $\mathbb{N}$ is finite. Then there exists a bijection $f : J_n \rightarrow \mathbb{N}$ for some $n \in \mathbb{N}$. Then, by [Theorem 2.3.2](#), $f(J_n) \subseteq \mathbb{N}$ is finite. By [Theorem 2.3.5](#), $f(J_n)$ has a greatest element, say m .

Then we have $m+1 \in \mathbb{N}$ and $m < m+1 \notin f(J_n)$, which contradicts the assumption that f is surjective. Therefore, $\mathbb{N}$ is infinite. $\square$

Theorem 2.4.4. *Any countable set is infinite.*

Proof. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \rightarrow X$. By [Theorem 2.4.2](#) and [lemma 2.4.3](#), X is infinite. $\square$

Remark. Let X be a countable set. Then there exists a bijection $f : \mathbb{N} \rightarrow X$. The elements of X can be listed as $X = \{x_1, x_2, x_3, \dots\}$, where $x_n = f(n)$ for all $n \in \mathbb{N}$. In other words, the elements of X can be arranged in a sequence.

Theorem 2.4.5. *$\mathbb{Z}$ is countable.*

Proof. Let $f : \mathbb{N} \rightarrow \mathbb{Z}$ be a function defined by

$$f(n) = \begin{cases} -\frac{n}{2}, & \text{if } n \text{ is even} \\ \frac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

We will show that f is bijective, which implies that $\mathbb{Z}$ is countable.

- **Injectivity:** Assume that $f(n_1) = f(n_2)$ for some $n_1, n_2 \in \mathbb{N}$. Since $f(n) \geq 0$ for odd n and $f(n) < 0$ for even n , we have n_1 and n_2 are either both even or both odd.

- If both n_1 and n_2 are even, then we have $-\frac{n_1}{2} = -\frac{n_2}{2}$, which implies $n_1 = n_2$.
- If both n_1 and n_2 are odd, then we have $\frac{n_1 - 1}{2} = \frac{n_2 - 1}{2}$, which implies $n_1 = n_2$.

Thus f is injective.

- **Surjectivity:** Let $k \in \mathbb{Z}$ be given. We need to find an $n \in \mathbb{N}$ such that $f(n) = k$.

- If $k \geq 0$, let $n = 2k + 1$. Then we have $f(n) = \frac{n - 1}{2} = \frac{2k}{2} = k$.
- If $k < 0$, let $n = -2k$. Then we have $f(n) = -\frac{n}{2} = -\frac{-2k}{2} = k$.

Thus f is surjective.

□

Lemma 2.4.6. *Any infinite subset of $\mathbb{N}$ is countable.*

Proof. Let $X \subseteq \mathbb{N}$ be an infinite subset. Let $\{X_n\}_{n=1}^{\infty}$ be a sequence and let $f : \mathbb{N} \rightarrow X$ be a function defined as follows:

- $X_1 = X, f(1) = \min X_1$
- $X_2 = X_1 \setminus \{f(1)\}, f(2) = \min X_2$
- $\vdots$
- $X_{n+1} = X_n \setminus \{f(n)\}, f(n+1) = \min X_{n+1}$

for each $n \in \mathbb{N}$. By [Theorem 2.4.1](#), X_n is infinite for each $n \in \mathbb{N}$, so is nonempty. Then, by [Theorem 1.1.1](#), X_n has a least element, so f is well-defined. We will show that f is bijective, which implies that X is countable.

- **Injectivity:** Assume that $n_1 \neq n_2$ for some $n_1, n_2 \in \mathbb{N}$. Without loss of generality, assume that $n_1 < n_2$. Then we have

$$f(n_1) \in X_{n_1} \text{ and } f(n_2) \in X_{n_2}.$$

Since $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$, we have $f(n_2) \neq f(n_1)$. Thus f is injective.

- **Surjectivity:** Let $x \in X$ be given. Let S be a set defined as

$$S := \{n \in \mathbb{N} \mid f(n) < x\}.$$

If $S = \emptyset$, then $f(1) = x$. Otherwise, by [Lemma 2.3.1](#), $S \subseteq \mathbb{N}$ is finite. Let $k = |S|$, $S = \{s_1, s_2, \dots, s_k\}$ with $s_1 < s_2 < \dots < s_k$. Then we have

- $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}.$
 - $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}.$
 - $\vdots$
 - $\min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} \implies X_{k+1} = X \setminus S.$
- Thus $f(k+1) = \min X_{k+1} = x$. Therefore, f is surjective.

□

Theorem 2.4.7. *Any infinite subset of a countable set is countable.*

Proof. Let X be a countable set and let $Y \subseteq X$ be an infinite subset. Since X is countable, there exists a bijection $f : \mathbb{N} \rightarrow X$. By Theorem 2.2.5, $f|_{f^{-1}(Y)}^Y : f^{-1}(Y) \rightarrow Y$ is bijective. Then, by Theorem 2.3.4, $f^{-1}(Y) \subseteq \mathbb{N}$ is infinite.

By Lemma 2.4.6, $f^{-1}(Y)$ is countable. Then there exists a bijection $g : \mathbb{N} \rightarrow f^{-1}(Y)$. The composition $f|_{f^{-1}(Y)}^Y \circ g : \mathbb{N} \rightarrow Y$ is bijective, which implies that Y is countable. □

Theorem 2.4.8. *Let X be infinite. If there exists a surjection $f : \mathbb{N} \rightarrow X$, then X is countable.*

Proof. For each $k \in X$, let S_k be a set defined as

$$S_k := f^{-1}(\{k\}).$$

Since f is surjective, $S_k \neq \emptyset$ and by Theorem 1.1.1, S_k has a least element for all $k \in X$. Let $g : X \rightarrow \mathbb{N}$ be a function defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Assume that $g(k_1) = g(k_2)$ for some $k_1, k_2 \in X$, i.e., $\min S_{k_1} = \min S_{k_2}$. Since $S_i \cap S_j = \emptyset$ for all $i, j \in X$ with $i \neq j$, we have $S_{k_1} = S_{k_2} \implies k_1 = k_2$. Thus g is injective.

Since g is injective, by Theorem 2.2.5, $g|_{g(X)} : X \rightarrow g(X)$ is bijective. Since X is infinite, $g(X) \subseteq \mathbb{N}$ is infinite by Theorem 2.4.2. Then, by Lemma 2.4.6, $g(X)$ is countable, which implies that X is countable. □

Theorem 2.4.9. *Let $\{X_n\}_{n \in \mathbb{N}}$ be a sequence of countable sets. Then $\bigcup_{n=1}^{\infty} X_n$ is countable, i.e., any countable union of countable sets is countable.*

Proof. There exists a bijection $f_n : \mathbb{N} \rightarrow X_n$ for each $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, we define the followings:

- $t_n := \min T_n$, where $T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\}$.
- $s_n := n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$.

Since $2n \in T_n$, T_n is nonempty. Then, by [Theorem 1.1.1](#), T_n has a least element, so t_n is well-defined. Since $\frac{t_n(t_n - 1)}{2} < n \leq \frac{t_n(t_n + 1)}{2}$, we have

$$t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1,$$

so $t_n - s_n + 1 \in \mathbb{N}$. Let $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$ be a function defined by

$$g(k) = f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let $x \in \bigcup_{n=1}^{\infty} X_n$ be given. Then there exists some $m \in \mathbb{N}$ such that $x \in X_m$. Since f_m is surjective, there exists some $l \in \mathbb{N}$ such that $f_m(l) = x$. Let

$$k := \frac{(l + m - 1)(l + m - 2)}{2} + m \in \mathbb{N}.$$

We will show that $l + m - 2 \notin T_k$ and $l + m - 1 \in T_k$, which implies that

- $t_k = \min T_k = l + m - 1$.
- $s_k = k - \frac{t_k(t_k - 1)}{2} = m$.
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l + m - 1 - m + 1) = f_m(l) = x$.

Consider the following inequalities when $z = l + m - 2$ and $z = l + m - 1$:

- Let $z = l + m - 2$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$

- Let $z = l + m - 1$. Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1 - l \leq 0. \end{aligned}$$

Thus g is surjective, which implies that $\bigcup_{n=1}^{\infty} X_n$ is countable by [Theorem 2.4.8](#). $\square$

Corollary 2.4.10. *Let $X_1, X_2, \dots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\bigcup_{i=1}^n X_i$ is countable, i.e., any finite union of countable sets is countable.*

Proof. Let $X_{n+1}, X_{n+2}, \dots$ be countable sets. Then, by [Theorem 2.4.9](#), $\bigcup_{n=1}^{\infty} X_n$ is countable. Assume that $\bigcup_{i=1}^n X_i$ is finite. Then $X_1 \subseteq \bigcup_{i=1}^n X_i$ is finite by [Theorem 2.3.2](#), which contradicts [Theorem 2.4.4](#). Thus $\bigcup_{i=1}^n X_i$ is infinite. Since $\bigcup_{i=1}^n X_i \subseteq \bigcup_{n=1}^{\infty} X_n$, by [Theorem 2.4.7](#), $\bigcup_{i=1}^n X_i$ is countable. $\square$

Theorem 2.4.11. *Let X be a countable set. If Y_x is finite for each $x \in X$, then the union*

$$Y = \bigcup_{x \in X} Y_x$$

is countable, i.e., any countable union of finite sets is countable.

Proof. There exists a bijection $f : \mathbb{N} \rightarrow X$. Also, for each $x \in X$, there exists a bijection $g_x : J_{n_x} \rightarrow Y_x$ for some $n_x \in \mathbb{N}$. Let $h : \mathbb{N} \rightarrow Y$ be a function defined by

- $h(j) = g_{f(1)}(j)$ for $j = 1, 2, \dots, n_{f(1)}$.
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) = g_{f(k)}(m)$ for each $k \in \mathbb{N}$ with $k \geq 2$, and $m = 1, 2, \dots, n_{f(k)}$.

Then h is surjective. Therefore, by [Theorem 2.4.8](#), Y is countable. $\square$

Lemma 2.4.12. $\mathbb{N} \times \mathbb{N}$ is countable.

Proof. We will prove this lemma by explicitly constructing a bijection from $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Let $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ be a function defined by

$$f(m, n) = \frac{(m + n - 1)(m + n - 2)}{2} + m$$

for each $(m, n) \in \mathbb{N} \times \mathbb{N}$. We will show that f is bijective.

- **Injectivity:** Assume that $(m_1, n_1) \neq (m_2, n_2)$. If $m_1 + n_1 = m_2 + n_2$, then $m_1 \neq m_2$, so $f(m_1, n_1) \neq f(m_2, n_2)$. Otherwise, $m_1 + n_1 \neq m_2 + n_2$.

Without loss of generality, assume $m_1 + n_1 < m_2 + n_2$. Then, since $m_1 + n_1 + 1 \leq m_2 + n_2$, we have

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2), \end{aligned}$$

Thus $f(m_1, n_1) \neq f(m_2, n_2)$, so f is injective.

- **Surjectivity:** Let $k \in \mathbb{N}$ be given. Let t_k, s_k be defined as in the proof of [Theorem 2.4.9](#). Then we have $f(s_k, t_k - s_k + 1) = k$. Thus f is surjective. $\square$

Theorem 2.4.13. Let X and Y be countable sets. Then the cartesian product $X \times Y$ is countable.

Proof. There exist bijections $f : \mathbb{N} \rightarrow X$ and $g : \mathbb{N} \rightarrow Y$. Let $h : \mathbb{N} \times \mathbb{N} \rightarrow X \times Y$ be a function defined by

$$h(m, n) = (f(m), g(n)) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}.$$

Then h is bijective. By [Lemma 2.4.12](#), $X \times Y$ is countable. $\square$

Theorem 2.4.14. Let $X_1, X_2, \dots, X_n$ be countable sets for some $n \in \mathbb{N}$. Then $\prod_{i=1}^n X_i$ is countable, i.e., any finite product of countable sets is countable.

Proof. Let $P(n)$ be the statement that

$$X_1, X_2, \dots, X_n \text{ are countable sets} \implies \prod_{i=1}^n X_i \text{ is countable.}$$

- **Base case:** Let X_1 be a countable set. Then $\prod_{i=1}^1 X_i = X_1$ is countable. Thus $P(1)$ is true.

- **Inductive step:** Assume that $P(n)$ is true for some $n \in \mathbb{N}$.

Let $X_1, X_2, \dots, X_{n+1}$ be countable sets. Then we have

$$\prod_{i=1}^{n+1} X_i = \left(\prod_{i=1}^n X_i \right) \times X_{n+1}$$

By the inductive hypothesis, $\prod_{i=1}^n X_i$ is countable. Then, by [Theorem 2.4.13](#),

$\prod_{i=1}^{n+1} X_i$ is countable. Thus $P(n+1)$ is true when $P(n)$ is true.

By the principle of mathematical induction, $P(n)$ is true for all $n \in \mathbb{N}$. □

Theorem 2.4.15. $\mathbb{Q}$ is countable.

Proof. Any $q \in \mathbb{Q}$ can be expressed as $q = \frac{m}{n}$ for some $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Let $f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$ be a function defined by

$$f(m, n) = \frac{m}{n} \text{ for all } (m, n) \in \mathbb{Z} \times \mathbb{N}.$$

Then f is surjective. By [Theorems 2.4.5](#) and [2.4.13](#), $\mathbb{Z} \times \mathbb{N}$ is countable. Then there exists a bijection $g : \mathbb{N} \rightarrow \mathbb{Z} \times \mathbb{N}$.

By [Theorem 2.2.7](#), $f \circ g : \mathbb{N} \rightarrow \mathbb{Q}$ is surjective. Since $\mathbb{N} \subseteq \mathbb{Q}$, by [Lemma 2.4.3](#) and [theorem 2.3.2](#), $\mathbb{Q}$ is infinite. Therefore, by [Theorem 2.4.8](#), $\mathbb{Q}$ is countable. □

Lemma 2.4.16. $X := \prod_{n=1}^{\infty} \{0, 1\}$ is uncountable.

Proof. For each $n \in \mathbb{N}$, let $e_n \in X$ be a sequence defined by

$$e_{nm} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{Otherwise} \end{cases}$$

Let $f : \mathbb{N} \rightarrow X$ be a function defined by

$$f(n) = e_n \text{ for all } n \in \mathbb{N}.$$

Then f is injective. By [Lemma 2.4.3](#) and [theorem 2.4.2](#), X is infinite.

Assume that X is countable. Then, there exists a bijection $g : \mathbb{N} \rightarrow X$. Let $X = \{g(1), g(2), g(3), \dots\}$, where

- $g(1) = (a_{11}, a_{12}, a_{13}, \dots)$.
- $g(2) = (a_{21}, a_{22}, a_{23}, \dots)$.
- $\vdots$

Let $x = (b_1, b_2, b_3, \dots) \in X$ be a sequence defined by

$$b_n = 1 - a_{nn} \text{ for all } n \in \mathbb{N}.$$

Then, $x \neq g(n)$ for all $n \in \mathbb{N}$, which contradicts the assumption that g is surjective. Therefore, X is uncountable. $\square$

Theorem 2.4.17. $\mathbb{R}$ is uncountable.

Proof. Since $\mathbb{N} \subseteq \mathbb{R}$, by [Lemma 2.4.3](#) and [theorem 2.3.2](#), $\mathbb{R}$ is infinite. Let $f : \mathbb{N} \rightarrow [0, 1)$ be a function defined by

$$f(n) = \frac{1}{n+1} \text{ for all } n \in \mathbb{N}.$$

Then f is injective. By [Lemma 2.4.3](#) and [theorem 2.4.2](#), $[0, 1)$ is infinite. We will prove that $[0, 1) \subseteq \mathbb{R}$ is uncountable, which implies that $\mathbb{R}$ is uncountable by [Theorem 2.4.7](#).

Let $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$ be a function defined by

$$g(x) = \{a_n^x\}_{n=1}^{\infty} \text{ for all } x \in [0, 1)$$

where $\{a_n^x\}_{n=1}^{\infty}$ is the greedy base 2 expansion of x . Let Y be a set defined by

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}.$$

Then, by [Theorems 1.4.9](#) and [2.2.5](#), $g|_Y$ is bijective.

Assume that $[0, 1)$ is countable. Then there exists a bijection $h : \mathbb{N} \rightarrow [0, 1)$. By [Theorem 2.2.7](#), $g|_Y \circ h : \mathbb{N} \rightarrow Y$ is bijective, so Y is countable. Let $Z := \prod_{n=1}^{\infty} \{0, 1\} \setminus Y$. By [Theorems 1.4.10](#) and [1.4.11](#), we have

$$Z = \left\{ \{a_n\}_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \text{ for some } N \in \mathbb{N} \right\}.$$

Let Z_N be a set defined by

$$Z_N := \left\{ \{a_n\}_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \right\} \text{ for each } N \in \mathbb{N}.$$

Let $F_N : \prod_{n=1}^{N-1} \{0, 1\} \rightarrow Z_N$ be a function defined by

$$F_N \left(\{a_n\}_{n=1}^{N-1} \right) = \{a_n\}_{n=1}^{N-1} \cup \{1\}_{n=N}^{\infty} \text{ for all } \{a_n\}_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0, 1\}.$$

Then F_N is surjective. By [Theorems 2.3.4](#) and [2.3.13](#), Z_N is finite. Then, by [Theorem 2.4.11](#), $Z = \bigcup_{N=1}^{\infty} Z_N$ is countable.

By [Corollary 2.4.10](#), $\prod_{n=1}^{\infty} \{0, 1\} = Y \cup Z$ is countable. [Lemma 2.4.16](#) shows that this is a contradiction. Therefore, $[0, 1)$ is uncountable. $\square$

2.5 Metric Spaces

Let X be a set. A function $d : X \times X \rightarrow \mathbb{R}$ is called a **metric** on X if for any $p, q \in X$, we have

(A) $d(p, q) \geq 0$ and $d(p, q) = 0 \iff p = q$.

(B) $d(p, q) = d(q, p)$.

(C) $d(p, r) \leq d(p, q) + d(q, r)$ for any $r \in X$.

(C) is called the **triangle inequality**. The pair (X, d) is called a **metric space**.

Theorem 2.5.1. For some $n \in \mathbb{N}$, let $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a function defined by

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then d is a metric on $\mathbb{R}^n$. Especially, d is called the **standard metric** on $\mathbb{R}^n$.

Proof. Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ be given.

- $d(\mathbf{x}, \mathbf{y}) \geq 0$ is trivial. Also, we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) = 0 &\iff \|\mathbf{x} - \mathbf{y}\| = 0 \\ &\iff x_k = y_k \text{ for all } k = 1, 2, \dots, n \\ &\iff \mathbf{x} = \mathbf{y}. \end{aligned}$$

- $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$ is trivial.
- By Theorem 1.7.2, we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| = \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \\ &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| = d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \end{aligned}$$

Therefore, d is a metric on $\mathbb{R}^n$. □

Definition 2.5.2. Let $k \in \mathbb{N}$ be given. A set $E \subseteq \mathbb{R}^k$ defined by

$$E := \left\{ \mathbf{x} = (x_1, x_2, \dots, x_k) \in \mathbb{R}^k \mid a_j \leq x_j \leq b_j \text{ for all } j = 1, 2, \dots, k \right\}$$

is called an **k -cell**, where $-\infty < a_j \leq b_j < +\infty$ for all $j = 1, 2, \dots, k$.

Definition 2.5.3. Let (X, d) be a metric space. For any $p \in X$ and $r \in \mathbb{R}^+$, we define the **neighborhood** of p with radius r as

$$N_r(p) := \{q \in X \mid d(p, q) < r\}.$$

Also, we define the **punctured neighborhood** of p with radius r as

$$Q_r(p) := N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}.$$

Especially, if $X = \mathbb{R}^n$ and d is the standard metric on $\mathbb{R}^n$ for some $n \in \mathbb{N}$, then $N_r(p)$ is called the **open ball** of radius r centered at p , which is denoted by $B_r(p)$.

Definition 2.5.4. Let (X, d) be a metric space and $E \subseteq X$. Let $p \in X$ be given.

- p is called an **limit point** of E if for any $r \in \mathbb{R}^+$, we have $E \cap Q_r(p) \neq \emptyset$.
- p is called a **isolated point** of E if there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \cap E = \{p\}$.
- p is called an **interior point** of E if there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \subseteq E$.

Definition 2.5.5. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **open** if any point of E is an interior point of E .
- E is said to be **closed** if any limit point of E belongs to E .

Definition 2.5.6. Let (X, d) be a metric space and $E \subseteq X$.

- E is said to be **bounded** if there exists some $p \in X$ and $r \in \mathbb{R}^+$ such that $E \subseteq N_r(p)$.
- E is said to be **perfect** if any point of E is a limit point of E .
- E is said to be **dense** if any point of X is either a limit point of E or a point of E .

Theorem 2.5.7. Let (X, d) be a metric space. For any $p \in X$ and $r \in \mathbb{R}^+$, the neighborhood $N_r(p)$ is open.

Proof. Let $q \in N_r(p)$ be given. Then we have $d(p, q) < r \implies \varepsilon := r - d(p, q) > 0$. Let $s \in N_\varepsilon(q)$ be given. Then we have

$$d(q, s) < \varepsilon \implies d(p, s) \leq d(p, q) + d(q, s) < d(p, q) + \varepsilon = r.$$

Thus $s \in N_r(p)$, which implies that $N_\varepsilon(q) \subseteq N_r(p)$. Since q is arbitrary, $N_r(p)$ is open. $\square$

Theorem 2.5.8. Let (X, d) be a metric space and $E \subseteq X$. Let p be a limit point of E . Then $E \cap N_r(p)$ is infinite for all $r \in \mathbb{R}^+$.

Proof. Assume that there exists some $r \in \mathbb{R}^+$ such that $E \cap N_r(p)$ is finite. Let S be a set defined by

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap Q_r(p).$$

Since p is a limit point of E , we have $S \neq \emptyset$. Since $E \cap N_r(p)$ is finite, $S \subseteq E \cap N_r(p)$ is also finite by [Theorem 2.3.2](#). Let T be a set defined by

$$T := \{d(p, q) \mid q \in S\} \neq \emptyset.$$

Since $p \notin S$, we have $0 \notin T$. Let $f : S \rightarrow T$ be a function defined by

$$f(q) = d(p, q).$$

Then f is surjective. By [Theorem 2.3.4](#), T is finite. Then, by [Theorem 2.3.5](#), T has a minimum element, say $\varepsilon = \min T > 0$. Then we have

- $N_\varepsilon(p) \subset N_r(p)$.
- $E \cap Q_\varepsilon(p) = \emptyset$.

This contradicts the assumption that p is a limit point of E . Therefore, $E \cap N_r(p)$ is infinite. $\square$

Theorem 2.5.9. Let (X, d) be a metric space and $E \subseteq X$. Then we have

$$E \text{ is open} \iff E^c \text{ is closed.}$$

Proof. ($\implies$) Assume that E is open. Let p be a limit point of E^c . Then, for any $r \in \mathbb{R}^+$, we have

$$E^c \cap Q_r(p) \neq \emptyset.$$

If $p \in E$, then there exists some $\varepsilon \in \mathbb{R}^+$ such that $N_\varepsilon(p) \subseteq E$. Then we have

$$E^c \cap Q_\varepsilon(p) = \emptyset,$$

which is a contradiction. Therefore, we have $p \in E^c$, which implies that E^c is closed.

($\impliedby$) Assume that E^c is closed. Let p be a point of E . If p is not an interior point of E , then for any $r \in \mathbb{R}^+$, we have

$$E^c \cap N_r(p) \neq \emptyset.$$

Since $p \in E \implies p \notin E^c$, we have

$$E^c \cap Q_r(p) \neq \emptyset,$$

which implies that p is a limit point of E^c . Since E^c is closed, we have $p \in E^c$, which is a contradiction. Therefore, p is an interior point of E . Since p is arbitrary, E is open. $\square$

Corollary 2.5.10. *Let (X, d) be a metric space and $E \subseteq X$. Then we have*

$$E \text{ is closed} \iff E^c \text{ is open.}$$

Theorem 2.5.11. *Let (X, d) be a metric space. Let $E_y \subseteq X$ be an open set for each $y \in Y$. Then we have*

$$\bigcup_{y \in Y} E_y \text{ is open}$$

whether Y is finite, countable, or uncountable. This theorem states that an arbitrary union of open sets is open.

Proof. Let $p \in \bigcup_{y \in Y} E_y$ be given. Then there exists some $y_0 \in Y$ such that $p \in E_{y_0}$. Since E_{y_0} is open, there exists some $r \in \mathbb{R}^+$ such that $N_r(p) \subseteq E_{y_0}$. Then we have

$$N_r(p) \subseteq E_{y_0} \subseteq \bigcup_{y \in Y} E_y,$$

which implies that p is an interior point of $\bigcup_{y \in Y} E_y$. Since p is arbitrary, $\bigcup_{y \in Y} E_y$ is open. $\square$

Corollary 2.5.12. *Let (X, d) be a metric space. Let $E_y \subseteq X$ be a closed set for each $y \in Y$. Then we have*

$$\bigcap_{y \in Y} E_y \text{ is closed}$$

whether Y is finite, countable, or uncountable. This corollary states that an arbitrary intersection of closed sets is closed.

Proof. By [Corollary 2.5.10](#), $(E_y)^c$ is open for each $y \in Y$. Then, by [Theorem 2.5.11](#), we have

$$\bigcup_{y \in Y} (E_y)^c \text{ is open.}$$

By [Theorem 2.5.9](#), we have

$$\bigcap_{y \in Y} E_y = \left(\bigcup_{y \in Y} (E_y)^c \right)^c \text{ is closed.}$$

$\square$

Theorem 2.5.13. *Let (X, d) be a metric space. Let $E_1, E_2, \dots, E_n \subseteq X$ be open sets for some $n \in \mathbb{N}$. Then we have*

$$\bigcap_{k=1}^n E_k \text{ is open.}$$

This theorem states that a finite intersection of open sets is open.

Proof. Let $p \in \bigcap_{k=1}^n E_k$ be given. Then we have $p \in E_j$ for all $j = 1, 2, \dots, n$. Since E_j is open, there exists some $r_j \in \mathbb{R}^+$ such that $N_{r_j}(p) \subseteq E_j$ for each $j = 1, 2, \dots, n$. Let R be a set defined by

$$R := \{r_j \mid j = 1, 2, \dots, n\} \subseteq \mathbb{R}^+.$$

Let $f : J_n \rightarrow R$ be a function defined by

$$f(j) = r_j.$$

Then f is surjective. By [Lemma 2.3.1](#) and [theorem 2.3.4](#), R is finite. Then, by [Theorem 2.3.5](#), R has a minimum element, say $r = \min R > 0$. Then we have

$$N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \text{ for all } j = 1, 2, \dots, n \implies N_r(p) \subseteq \bigcap_{k=1}^n E_k,$$

which implies that p is an interior point of $\bigcap_{k=1}^n E_k$. Since p is arbitrary, $\bigcap_{k=1}^n E_k$ is open. $\square$

Corollary 2.5.14. *Let (X, d) be a metric space. Let $E_1, E_2, \dots, E_n \subseteq X$ be closed sets for some $n \in \mathbb{N}$. Then we have*

$$\bigcup_{k=1}^n E_k \text{ is closed.}$$

This corollary states that a finite union of closed sets is closed.

Proof. By [Corollary 2.5.10](#), $(E_k)^c$ is open for each $k = 1, 2, \dots, n$. Then, by [Theorem 2.5.13](#), we have

$$\bigcap_{k=1}^n (E_k)^c \text{ is open.}$$

By [Theorem 2.5.9](#), we have

$$\bigcup_{k=1}^n E_k = \left(\bigcap_{k=1}^n (E_k)^c \right)^c \text{ is closed.}$$

$\square$

Definition 2.5.15. Let (X, d) be a metric space and $E \subseteq X$. Let E' be the set of all limit points of E . The **closure** of E , which is denoted by $\overline{E}$, is defined by

$$\overline{E} := E \cup E'.$$

Theorem 2.5.16. Let (X, d) be a metric space and $E \subseteq X$. Then we have the following properties:

(A) $\overline{E}$ is closed.

(B) $E = \overline{E} \iff E$ is closed.

(C) For any closed subset $F \subseteq X$ such that $E \subseteq F$, we have $\overline{E} \subseteq F$.

(C) states that the closure of E is the smallest closed set containing E .

Proof. (A) We will show that $(\overline{E})^c$ is open. Let $p \in (\overline{E})^c$ be given. Since $p \notin E$ and $p \notin E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap Q_r(p) = \emptyset \implies E \cap N_r(p) = \emptyset \implies N_r(p) \subseteq E^c.$$

By Theorem 2.5.7, $N_r(p)$ is open. Then, for any $q \in N_r(p)$, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$\begin{aligned} N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c &\implies E \cap N_\varepsilon(q) = \emptyset \\ &\implies E \cap (Q_\varepsilon(q) \cup \{q\}) = \emptyset \\ &\implies (E \cap Q_\varepsilon(q)) \cup (E \cap \{q\}) = \emptyset \\ &\implies E \cap Q_\varepsilon(q) = \emptyset \\ &\implies q \notin E'. \end{aligned}$$

Since q is arbitrary, we have

$$N_r(p) \subseteq (E')^c \implies N_r(p) \subseteq (\overline{E})^c = E^c \cap (E')^c,$$

which implies that p is an interior point of $(\overline{E})^c$. Since p is arbitrary, $(\overline{E})^c$ is open, which implies that $\overline{E}$ is closed by Corollary 2.5.10.

(B) E is closed $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$.

(C) Let $F \subseteq X$ be a closed set such that $E \subseteq F$. Let $p \in E'$ be given. Then, for any $r \in \mathbb{R}^+$, we have

$$E \cap Q_r(p) \neq \emptyset \implies F \cap Q_r(p) \neq \emptyset,$$

which implies that $p \in F' \subseteq F$. Therefore, we have $E' \subseteq F$. Since $E \subseteq F$, we have $\overline{E} = E \cup E' \subseteq F$.

□

Theorem 2.5.17. *Let (X, d) be a metric space and $E \subseteq X$. Then we have*

$$E \setminus E' = \{p \in X \mid p \text{ is an isolated point of } E\}.$$

Proof. ($\subseteq$) Let $p \in E \setminus E'$ be given. Since $p \notin E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap Q_r(p) = \emptyset \implies (E \cap N_r(p)) \setminus \{p\} = \emptyset.$$

Since $p \in E$ and $p \in N_r(p)$, we have

$$E \cap N_r(p) = \{p\},$$

which implies that p is an isolated point of E .

($\supseteq$) Let p be an isolated point of E . Then there exists some $r \in \mathbb{R}^+$ such that

$$E \cap N_r(p) = \{p\}.$$

Then we have

- $E \cap Q_r(p) = \emptyset \implies p \notin E'$.
- $p \in E$.

Therefore, we have $p \in E \setminus E'$. □

Definition 2.5.18. Let (X, d) be a metric space and $E \subseteq X$. The **boundary** of E , which is denoted by ∂E , is defined by

$$\partial E := \overline{E} \cap \overline{E^c}.$$

Lemma 2.5.19. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for some $n \in \mathbb{N}$. Then we have*

$$Q_r(\mathbf{p}) \neq \emptyset \quad \text{for all } \mathbf{p} \in \mathbb{R}^n \text{ and } r \in \mathbb{R}^+.$$

Proof. Let $\mathbf{p} = (p_1, p_2, \dots, p_n) \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ be given. Then we have

$$d\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1, \mathbf{p}\right) = \left\|\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) - \mathbf{p}\right\| = \left\|\frac{r}{2}\mathbf{e}_1\right\| = \frac{r}{2}\|\mathbf{e}_1\| = \frac{r}{2} < r.$$

where $\mathbf{e}_1 = (1, 0, 0, \dots, 0) \in \mathbb{R}^n$. Therefore, we have

$$\mathbf{p} \neq \left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) \in Q_r(\mathbf{p}) \implies Q_r(\mathbf{p}) \neq \emptyset.$$

□

Theorem 2.5.20. *Let $(\mathbb{R}^n, \|\cdot\|)$ be given for some $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have*

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)'),$$

which states that the boundary of E consists the following three disjoint sets:

- *The set of all isolated points of E .*
- *The set of all isolated points of E^c .*
- *The set of all limit points of both E and E^c .*

Proof. By the definition of the boundary, we have

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= (E \cup E') \cap (E^c \cup (E^c)') \\ &= (E \cap E^c) \cup (E \cap (E^c)') \cup (E' \cap E^c) \cup (E' \cap (E^c)'). \end{aligned}$$

Consider the following equivalences:

- $E = (E \setminus E') \cup (E \cap E')$.
- $E^c = (E^c \setminus (E^c)') \cup (E^c \cap (E^c)').$

Then we have the following equalities:

- $E \cap (E^c)' = ((E \setminus E') \cap (E^c)') \cup ((E \cap E') \cap (E^c)').$
- $E' \cap E^c = (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c \cap (E^c))).$

Consider the last two terms of the above equalities:

- $((E \cap E') \cap (E^c)') \subseteq E' \cap (E^c)'.$
- $(E' \cap (E^c \cap (E^c))) \subseteq E' \cap (E^c)'.$

Then we have

$$\begin{aligned} \partial E &= \emptyset \cup ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \\ &= ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \end{aligned}$$

For any $\mathbf{p} \in E \setminus E'$, there exists some $r \in \mathbb{R}^+$ such that

$$E \cap Q_r(\mathbf{p}) = \emptyset \implies Q_r(\mathbf{p}) \subseteq E^c.$$

By [Lemma 2.5.19](#), we have

$$Q_s(\mathbf{p}) \cap Q_r(\mathbf{p}) = Q_{\min(s,r)}(\mathbf{p}) \neq \emptyset \quad \text{for all } s \in \mathbb{R}^+.$$

Then, for any $s \in \mathbb{R}^+$, we have

$$\begin{aligned} Q_s(\mathbf{p}) \cap Q_r(\mathbf{p}) \neq \emptyset &\implies Q_s(\mathbf{p}) \cap (Q_r(\mathbf{p}) \cap E^c) \neq \emptyset \\ &\implies \emptyset \neq (Q_s(\mathbf{p}) \cap E^c) \cap Q_r(\mathbf{p}) \subseteq Q_s(\mathbf{p}) \cap E^c \\ &\implies E^c \cap Q_s(\mathbf{p}) \neq \emptyset, \end{aligned}$$

which implies that $\mathbf{p} \in (E^c)'$. Then we have

$$E \setminus E' \subseteq (E^c)' \implies (E \setminus E') \cap (E^c)' = E \setminus E'.$$

Similarly, we can show that

$$E^c \setminus (E^c)' \subseteq E' \implies E' \cap (E^c \setminus (E^c)') = E^c \setminus (E^c)'.$$

Therefore, we have

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)').$$

□

Remark. Let (X, d) be a metric space and $Y \subseteq X$. Then, $(Y, d|_{Y \times Y})$ is also a metric space, since

$$d|_{Y \times Y} : Y \times Y \rightarrow \mathbb{R}$$

satisfies the properties of a metric. We call $(Y, d|_{Y \times Y})$ a **subspace** of (X, d) . For the sake of simplicity, we will denote $d|_{Y \times Y}$ by d and write (Y, d) instead of $(Y, d|_{Y \times Y})$.

Definition 2.5.21. Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. E is called **open relative** to Y if E is open in the subspace (Y, d) . That is, for any $p \in E$, there exists some $r \in \mathbb{R}^+$ such that

$$Y \cap N_r(p) \subseteq E.$$

Theorem 2.5.22. Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is open relative to Y if and only if there exists an open set V in X such that $E = V \cap Y$.

Proof. ($\implies$) Let E be an open set relative to Y . Then, for each $p \in E$, there exists some $r_p \in \mathbb{R}^+$ such that

$$Y \cap N_{r_p}(p) \subseteq E.$$

Let V be a set defined by

$$V := \bigcup_{p \in E} N_{r_p}(p).$$

By [Theorems 2.5.7](#) and [2.5.11](#), V is open in X . Since $p \in N_{r_p}(p) \subseteq V$ for each $p \in E$, we have

- $E \subseteq Y$ and $E \subseteq V \implies E \subseteq V \cap Y$.
- $\bigcup_{p \in E} (Y \cap N_{r_p}(p)) \subseteq E \implies Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \implies Y \cap V \subseteq E$.

Therefore, we have $E = V \cap Y$.

($\Leftarrow$) Let V be an open set in X such that $E = V \cap Y$. Then, for each $p \in E \subseteq V$, there exists some $r_p \in \mathbb{R}^+$ such that

$$N_{r_p}(p) \subseteq V \implies Y \cap N_{r_p}(p) \subseteq V \cap Y = E.$$

□

2.6 Compact Sets

Let (X, d) be a metric space and $E \subseteq X$. Let A be a set and let G_α be an open set in X for each $\alpha \in A$. Then the collection $\{G_\alpha\}_{\alpha \in A}$ is called an **open cover** of E if

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

A collection $\{G_\alpha\}_{\alpha \in S}$ is called a **subcover** of E if

- $S \subseteq A$.
- $E \subseteq \bigcup_{\alpha \in S} G_\alpha$.

E is called **compact** if any open cover of E has a finite subcover. That is, for each open cover $\{G_\alpha\}_{\alpha \in A}$ of E , there exists a subset $S_A \subseteq A$ such that

- S_A is finite.
- $E \subseteq \bigcup_{\alpha \in S_A} G_\alpha$.

Theorem 2.6.1. *Let (X, d) be a metric space and $E \subseteq Y \subseteq X$. Then E is compact relative to Y if and only if E is compact relative to X .*

Proof. ($\implies$) Let E be a compact set relative to Y . Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of E relative to X . Then we have

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Let $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. By [Theorem 2.5.22](#), H_α is open relative to Y . Since $E \subseteq Y$, we have

$$E \subseteq \left(\bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in A}$ is an open cover of E relative to Y . Since E is compact relative to Y , there exists a finite subcover $\{H_\alpha\}_{\alpha \in S_A}$ of E relative to Y . Then we have

$$E \subseteq \bigcup_{\alpha \in S_A} H_\alpha = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S_A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to X . Therefore, E is compact relative to X .

($\Leftarrow$) Let E be a compact set relative to X . Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of E relative to Y . Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha.$$

By [Theorem 2.5.22](#), there exists an open set G_α in X such that $H_\alpha = G_\alpha \cap Y$ for each $\alpha \in A$. Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Thus $\{G_\alpha\}_{\alpha \in A}$ is an open cover of E relative to X . Since E is compact relative to X , there exists a finite subcover $\{G_\alpha\}_{\alpha \in S_A}$ of E relative to X . Then we have

$$E \subseteq \left(\bigcup_{\alpha \in S_A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) = \bigcup_{\alpha \in S_A} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in S_A}$ is a finite subcover of E relative to Y . Therefore, E is compact relative to Y . $\square$

Remark. The concept of **compactness** is preserved under the subspace topology, while the concepts of **openness** and **closedness** are not, which is the reason why

“Let (X, d) be a metric space and $E \subseteq X$.”

this kind of statement is always included in theorems and definitions about metric spaces. In this textbook, you can assume that

- E is open. $\implies E$ is open relative to X .
- E is closed. $\implies E$ is closed relative to X .

- E is compact. $\implies E$ is compact relative to X .

if there is no specific mention of the subspace topology.

Theorem 2.6.2. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is closed.*

Proof. Let E be compact. We will prove that E^c is open, which implies that E is closed by [Corollary 2.5.10](#). Let $p \in E^c$ be given. Since $p \notin E$, we have

$$r_q := \frac{d(p, q)}{2} > 0 \quad \text{for each } q \in E.$$

Since $q \in N_{r_q}(q)$ for each $q \in E$, we have

$$E \subseteq \bigcup_{q \in E} N_{r_q}(q) \implies \{N_{r_q}(q)\}_{q \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{N_{r_q}(q)\}_{q \in S}$ of E . Let T be a set defined by

$$T := \{r_q \mid q \in S\}.$$

Let $f : S \rightarrow T$ be a function defined by

$$f(q) = r_q.$$

Then f is surjective. By [Theorem 2.3.4](#), T is finite. Then, by [Theorem 2.3.5](#), T has a minimum element, say $\varepsilon = \min T > 0$. For any $x \in N_\varepsilon(p)$, we have

$$d(x, p) < \varepsilon \leq r_q \quad \text{for each } q \in S.$$

Assume that there exists some $q \in S$ such that $x \in N_{r_q}(q)$. Then we have

$$d(p, q) \leq d(p, x) + d(x, q) < r_q + r_q = 2r_q = d(p, q),$$

which is a contradiction. Then the following statements hold:

- $N_\varepsilon(p) \cap N_{r_q}(q) = \emptyset$ for each $q \in S$.
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$.

Therefore, we have $N_\varepsilon(p) \subseteq E^c$. Since p is arbitrary, E^c is open. □

Theorem 2.6.3. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then E is bounded.*

Proof. Let E be compact. Let $p \in E$ be given. Let $\{G_n\}_{n \in \mathbb{N}}$ be a collection of open sets defined by

$$G_n = N_n(p) \quad \text{for each } n \in \mathbb{N}.$$

Then we have

$$n \leq m \implies G_n \subseteq G_m \quad \text{for each } n, m \in \mathbb{N}.$$

For any $q \in X$, by [Theorem 1.4.1](#), there exists $n \in \mathbb{N}$ such that $n > d(p, q)$. Thus we have

$$E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \implies \{G_n\}_{n \in \mathbb{N}} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_n\}_{n \in S}$ of E . Since $S \subseteq \mathbb{N}$ is finite, by [Theorem 2.3.5](#), S has a greatest element, say $m = \max S$. Then we have

$$E \subseteq \bigcup_{n \in S} G_n \subseteq G_m = N_m(p),$$

which implies that E is bounded. □

Theorem 2.6.4. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then any closed subset $F \subseteq E$ is compact.*

Proof. Let $\{H_\alpha\}_{\alpha \in A}$ be an open cover of F . We have the following statements:

- F is closed. $\implies F^c$ is open.
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$.

Then we have

- $X = F \cup F^c \subseteq \left(\bigcup_{\alpha \in A} H_\alpha \right) \cup F^c = \bigcup_{\alpha \in A} (H_\alpha \cup F^c)$.
- $E \subseteq X \implies E \subseteq \bigcup_{\alpha \in A} (H_\alpha \cup F^c)$.

By [Theorem 2.5.11](#), $H_\alpha \cup F^c$ is open for each $\alpha \in A$. Thus

$$\{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{H_\alpha \cup F^c\}_{\alpha \in B}$ of E . Then we have

- $E \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c)$.

$$\bullet F \subseteq E \implies F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left(\bigcup_{\alpha \in B} H_\alpha \right) \cup F^c.$$

Since $F \cap F^c = \emptyset$, we have

$$F \subseteq \bigcup_{\alpha \in B} H_\alpha.$$

Thus $\{H_\alpha\}_{\alpha \in B}$ is a finite subcover of F . Therefore, F is compact. $\square$

Theorem 2.6.5. *Let (X, d) be a metric space and $E \subseteq X$. If E is compact, then any infinite subset of E has a limit point in E .*

Proof. Let E be compact. Let $F \subseteq E$ be an infinite subset of E . Assume that F has no limit point in E . Then, for each $p \in E$, there exists $r_p \in \mathbb{R}^+$ such that

$$Q_{r_p}(p) \cap F = \emptyset \implies N_{r_p}(p) \cap F \subseteq \{p\}.$$

Let $\{G_\alpha\}_{\alpha \in E}$ be a collection of open sets defined by

$$G_\alpha = N_{r_\alpha}(\alpha) \quad \text{for each } \alpha \in E.$$

Since $\alpha \in G_\alpha$ for each $\alpha \in E$, we have

$$E \subseteq \bigcup_{\alpha \in E} G_\alpha \implies \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E.$$

Since E is compact, there exists a finite subcover $\{G_\alpha\}_{\alpha \in S}$ of E . Then we have

$$E \subseteq \bigcup_{\alpha \in S} G_\alpha \implies F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha.$$

Since $F \subseteq \bigcup_{\alpha \in S} G_\alpha$, we have

$$F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S,$$

which is a contradiction by [Theorem 2.3.2](#). Therefore, any infinite subset of E has a limit point in E . $\square$

Theorem 2.6.6. *Let (X, d) be a metric space and let $\{K_\alpha\}_{\alpha \in A}$ be a collection of compact sets. Then we have*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset \quad \text{for all finite subset } B \subseteq A \implies \bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

Proof. Assume that $\bigcap_{\alpha \in A} K_\alpha = \emptyset$. Then we have

$$X = \left(\bigcap_{\alpha \in A} K_\alpha \right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c.$$

Since each K_α is compact, K_α is closed by [Theorem 2.6.2](#). Then, by [Corollary 2.5.10](#), we have

Each $(K_\alpha)^c$ is open $\implies \{(K_\alpha)^c\}_{\alpha \in A}$ is an open cover of X .

Since $K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c$, we have

$\{(K_\beta)^c\}_{\beta \in A}$ is an open cover of K_α .

Since K_α is compact, there exists a finite subcover $\{(K_\beta)^c\}_{\beta \in B_\alpha}$ of K_α . Then we have

$$\begin{aligned} K_\alpha &\subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left(\bigcap_{\beta \in B_\alpha} K_\beta \right)^c \\ &\implies K_\alpha \cap \left(\bigcap_{\beta \in B_\alpha} K_\beta \right) = \emptyset \\ &\implies \bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta = \emptyset. \end{aligned}$$

Since $\{\alpha\}$ and B_α are finite, $B_\alpha \cup \{\alpha\}$ is also finite by [Theorem 2.3.11](#), which contradicts the assumption. Therefore, we have

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

□

Lemma 2.6.7. Let $\{a_n\}_{n=1}^\infty$ and $\{b_n\}_{n=1}^\infty$ be sequences in $\mathbb{R}$, and let $\{I_n\}_{n=1}^\infty$ be a sequence of intervals in $\mathbb{R}$ such that

- $I_n = [a_n, b_n]$ for all $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$

where $a_n \leq b_n$ for all $n \in \mathbb{N}$. Then we have

$$\bigcap_{n=1}^\infty I_n \neq \emptyset.$$

Proof. Let E be a set defined by

$$E := \{a_n \mid n \in \mathbb{N}\}.$$

Since E is bounded above by b_1 , there exists $\alpha = \sup E$. Then we have

$$a_m \leq \alpha \quad \text{for all } m \in \mathbb{N}.$$

For any $m \in \mathbb{N}$, we have b_m is an upper bound of E since

- $a_n \leq a_m \leq b_m$ for all $n \in \mathbb{N}$ such that $n \leq m$.
- $a_n \leq b_n \leq b_m$ for all $n \in \mathbb{N}$ such that $n > m$.

Therefore, we have

$$a_m \leq \alpha \leq b_m \implies \alpha \in I_m \text{ for all } m \in \mathbb{N}.$$

□

Theorem 2.6.8. Let $k \in \mathbb{N}$ be given. Let $\{a_{n,j}\}_{n=1}^{\infty}$ and $\{b_{n,j}\}_{n=1}^{\infty}$ be sequences in $\mathbb{R}$ for all $j = 1, 2, \dots, k$, and let $\{I_n\}_{n=1}^{\infty}$ be a sequence of k -cells in $\mathbb{R}^k$ such that

- $I_n = \{\mathbf{x} \in \mathbb{R}^k \mid a_{n,j} \leq x_j \leq b_{n,j} \text{ for all } j = 1, 2, \dots, k\}$ for all $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j} \text{ for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}$

where $a_{n,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$. Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

Proof. Let $\{I_{n,j}\}_{n=1}^{\infty}$ be a sequence of intervals in $\mathbb{R}$ defined by

$$I_{n,j} = [a_{n,j}, b_{n,j}] \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Then we have $I_{n+1,j} \subseteq I_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$. By [Lemma 2.6.7](#), we have

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \emptyset \quad \text{for all } j = 1, 2, \dots, k.$$

Let $\alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j}$ be given for all $j = 1, 2, \dots, k$. Then we have

$$\alpha_j \in I_{n,j} \implies a_{n,j} \leq \alpha_j \leq b_{n,j} \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Let $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k) \in \mathbb{R}^k$. Then we have

$$\alpha \in I_n \quad \text{for all } n \in \mathbb{N} \implies \alpha \in \bigcap_{n=1}^{\infty} I_n.$$

□

Theorem 2.6.9. Let $(\mathbb{R}^k, \|\cdot\|)$ be given for some $k \in \mathbb{N}$. Then any k -cell is compact.

Proof. Let I be a k -cell in $\mathbb{R}^k$, and let $I_1 := I$. Then we have

$$I_1 = \prod_{j=1}^k [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k,$$

where $-\infty < a_{1,j} \leq b_{1,j} < +\infty$ for all $j = 1, 2, \dots, k$. Let $\{I_{1,j,m}\}$ be an interval in $\mathbb{R}$ defined by

$$I_{1,j,m} := \begin{cases} \left[a_{1,j}, \frac{a_{1,j} + b_{1,j}}{2} \right] & \text{if } m = 1 \\ \left[\frac{a_{1,j} + b_{1,j}}{2}, b_{1,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Let S_k be a set defined by

$$S_k := \{1, 2\}^k = \left\{ \mathbf{m} \in \mathbb{R}^k \mid m_j \in \{1, 2\} \text{ for all } j = 1, 2, \dots, k \right\}.$$

By [Theorem 2.3.13](#), S_k is finite. We have

$$I_1 = \bigcup_{\mathbf{m} \in S_k} \left(\prod_{j=1}^k I_{1,j,m_j} \right).$$

Let $\{G_\alpha\}_{\alpha \in A}$ be an open cover of I_1 . Assume that I_1 has no finite subcover. Then, for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^k I_{1,j,m_j}$. Assume that for each $\mathbf{m} \in S_k$, we have

$$\prod_{j=1}^k I_{1,j,m_j} \text{ has a finite subcover } \{G_\alpha\}_{\alpha \in A_{\mathbf{m}}}.$$

Then we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \quad \text{for all } \mathbf{m} \in S_k,$$

where $\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha$ is open by [Theorem 2.5.11](#). Then we have

$$I_1 \subseteq \bigcup_{\mathbf{m} \in S_k} \left(\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \right) = \bigcup_{\alpha \in A} G_\alpha,$$

where $A := \bigcup_{m \in S_k} A_m$ is finite by [Theorem 2.3.11](#). Thus I_1 has a finite subcover, which is a contradiction. Therefore, there exists $m_1 \in S_k$ such that

$$\prod_{j=1}^k I_{1,j,(m_1)_j} \text{ has no finite subcover.}$$

Let I_2 be a k -cell in $\mathbb{R}^k$ defined by

$$I_2 := \prod_{j=1}^k [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k,$$

where $a_{2,j}, b_{2,j} \in \mathbb{R}$ is defined by

$$\begin{aligned} \bullet \quad a_{2,j} &:= \begin{cases} a_{1,j} & \text{if } (m_1)_j = 1 \\ \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 2 \end{cases} \\ \bullet \quad b_{2,j} &:= \begin{cases} \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 1 \\ b_{1,j} & \text{if } (m_1)_j = 2 \end{cases} \end{aligned}$$

for all $j = 1, 2, \dots, k$. Then, for each $j = 1, 2, \dots, k$, we have

$$[a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}],$$

which implies that

$$I_2 = \prod_{j=1}^k I_{1,j,(m_1)_j} \subseteq I_1.$$

Let $\{I_{2,j,m}\}$ be an interval in $\mathbb{R}$ defined by

$$I_{2,j,m} := \begin{cases} \left[a_{2,j}, \frac{a_{2,j} + b_{2,j}}{2} \right] & \text{if } m = 1 \\ \left[\frac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Then we have

$$I_2 = \bigcup_{m \in S_k} \left(\prod_{j=1}^k I_{2,j,m_j} \right).$$

We have shown that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of I_2 , and I_2 has no finite subcover. Then, for each $m \in S_k$, we have

$$\prod_{j=1}^k I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that $\{G_\alpha\}_{\alpha \in A}$ is an open cover of $\prod_{j=1}^k I_{2,j,m_j}$. Therefore, there exists $\mathbf{m}_2 \in S_k$ such that

$$\prod_{j=1}^k I_{2,j,(\mathbf{m}_2)_j} \text{ has no finite subcover.}$$

By repeating this process, we can construct the following sequences:

- The sequence $\{\mathbf{m}_n\}_{n=1}^\infty$ in S_k .
- The sequences $\{a_{n,j}\}_{n=1}^\infty$ and $\{b_{n,j}\}_{n=1}^\infty$ in $\mathbb{R}$ for all $j = 1, 2, \dots, k$.
- The sequence $\{I_{n,j,m}\}_{n=1}^\infty$ of intervals in $\mathbb{R}$ for all $j = 1, 2, \dots, k$ and $m = 1, 2$.
- The sequence $\{I_n\}_{n=1}^\infty$ of k -cells in $\mathbb{R}^k$.

We have the following properties:

- $I_{n+1} \subseteq I_n$ for all $n \in \mathbb{N}$.
- I_n has no finite subcover for all $n \in \mathbb{N}$.
- $a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$.
- $b_{n+1,j} - a_{n+1,j} = \frac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \frac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$ for all $j = 1, 2, \dots, k$ and $n \in \mathbb{N}$.

Let $\delta := \sqrt{\sum_{j=1}^k (b_{1,j} - a_{1,j})^2}$. Then we have

$$\|\mathbf{x} - \mathbf{y}\| \leq \sqrt{\sum_{j=1}^k (b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } \mathbf{x}, \mathbf{y} \in I_n \text{ and } n \in \mathbb{N}.$$

By [Theorem 2.6.8](#), we have $\bigcap_{n=1}^\infty I_n \neq \emptyset$. Let $\beta \in \bigcap_{n=1}^\infty I_n$ be given. Then there exists $\alpha_0 \in A$ such that $\beta \in G_{\alpha_0}$. Since G_{α_0} is open, there exists $\varepsilon \in \mathbb{R}^+$ such that

$$B_\varepsilon(\beta) \subseteq G_{\alpha_0}.$$

By [Corollary 1.4.2](#), there exists $N \in \mathbb{N}$ such that $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$. Since $\beta \in I_N$, we have

$$\|\beta - \gamma\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \gamma \in I_N,$$

which implies that $I_N \subseteq B_\varepsilon(\beta) \subseteq G_{\alpha_0}$. Therefore, I_N has a finite subcover $\{G_{\alpha_0}\}$, which is a contradiction. Thus I has a finite subcover, and I is compact. $\square$

Corollary 2.6.10. For any $-\infty < a \leq b < +\infty$, the closed interval $[a, b]$ is compact in $(\mathbb{R}, \|\cdot\|)$.

Theorem 2.6.11. Let $(\mathbb{R}^n, \|\cdot\|)$ be given for some $n \in \mathbb{N}$ and $E \subseteq \mathbb{R}^n$. Then we have

$$E \text{ is compact} \iff E \text{ is closed and bounded.}$$

This theorem is called the **Heine-Borel Theorem**.

Proof. ($\implies$) Let E be compact. E is closed by [Theorem 2.6.2](#), and E is bounded by [Theorem 2.6.3](#).

($\impliedby$) Let E be closed and bounded. Then there exists some $\mathbf{p} \in \mathbb{R}^n$ and $r \in \mathbb{R}^+$ such that $E \subseteq B_r(\mathbf{p})$. Let I be a n -cell in $\mathbb{R}^n$ defined by

$$I := \prod_{j=1}^n [p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have $E \subseteq B_r(\mathbf{p}) \subseteq I$. By [Theorem 2.6.9](#), I is compact. Since E is closed, by [Theorem 2.6.4](#), E is compact. $\square$

2.7 Connected Sets

Let (X, d) be a metric space and $E \subseteq X$. E is called **connected** if there exists no pair of sets $A, B \subseteq X$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Such pair of A and B is called **separated**. If E is not connected, then E is called **disconnected**.

Theorem 2.7.1. Let $(\mathbb{R}, \|\cdot\|)$ be given. Then $E \subseteq \mathbb{R}$ is connected if and only if

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Proof. ($\implies$) Let $E \subseteq \mathbb{R}$ be connected. Assume that there exist $a, b \in E$ and $x \in \mathbb{R}$ such that

$$a < x < b \text{ and } x \notin E.$$

Let A, B be sets defined by

- $A := E \cap (-\infty, x) \subseteq E$.
- $B := E \cap (x, \infty) \subseteq E$.

Then we have

- $A \neq \emptyset$ and $B \neq \emptyset$ since $a \in A$ and $b \in B$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$ since $x \notin E$.

Thus A and B are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

($\Leftarrow$) Let $E \subseteq \mathbb{R}$ be a set such that

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume that E is disconnected. Then there exist separated sets $A, B \subseteq X$ such that

- $\emptyset \neq A \subseteq E$ and $\emptyset \neq B \subseteq E$.
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$.
- $E = A \cup B$.

Let $a \in A$ and $b \in B$ be given. Without loss of generality, assume that $a < b$. Let

$$x := \sup(A \cap [a, b]),$$

whose *existence* is guaranteed by the [least upper bound property](#). □

Chapter 3

Numerical Sequences and Series

3.1 Convergent Sequences

3.2 Subsequences

3.3 Cauchy Sequences

3.4 Upper and Lower Limits

3.5 Some Special Sequences

3.6 Series

3.7 Series of Nonnegative Terms

3.8 The Number e

3.9 The Root and Ratio Tests

3.10 Power Series

3.11 Summation by Parts

3.12 Absolute Convergence

3.13 Addition and Multiplication of Series

3.14 Rearrangements

Chapter 4

Continuity

4.1 Limits of Functions

4.2 Continuous Functions

4.3 Continuity and Compactness

4.4 Continuity and Connectedness

4.5 Discontinuities

4.6 Monotonic Functions

4.7 Infinite Limits and Limits at Infinity

Chapter 5

Differentiation

5.1 The Derivative of a Real Function

5.2 Mean Value Theorems

5.3 The Continuity of Derivatives

5.4 L'Hospital's Rule

5.5 Derivatives of Higher Order

5.6 Taylor's Theorem

5.7 Differentiation of Vector-valued Functions

Chapter 6

The Riemann-Stieltjes Integral

6.1 Definition and Existence of the Integral

6.2 Properties of the Integral

6.3 Integration and Differentiation

6.4 Integration of Vector-valued Functions

6.5 Rectifiable Curves

Chapter 7

Sequences and Series of Functions

7.1 Discussion of Main Problem

7.2 Uniform Convergence

7.3 Uniform Convergence and Continuity

7.4 Uniform Convergence and Integration

7.5 Uniform Convergence and Differentiation

7.6 Equicontinuous Families of Functions

7.7 Equicontinuous Families of Functions

7.8 The Stone-Weierstrass Theorem

Chapter 8

Some Special Functions

8.1 Power Series

8.2 The Exponential and Logarithmic Functions

8.3 The Trigonometric Functions

8.4 The Algebraic Completeness of the Complex Field

8.5 Fourier Series

8.6 The Gamma Function

Chapter 9

Functions of Several Variables

9.1 Linear Transformations

9.2 Differentiation

9.3 The Contraction Principle

9.4 The Inverse Function Theorem

9.5 The Implicit Function Theorem

9.6 The Rank Theorem

9.7 Determinants

9.8 Derivatives of Higher Order

9.9 Differentiation of Integrals

Chapter 10

Integration of Differential Forms

10.1 Integration

10.2 Primitive Mappings

10.3 Partitions of Unity

10.4 Change of Variables

10.5 Differential Forms

10.6 Simplexes and Chains

10.7 Stokes' Theorem

10.8 Closed Forms and Exact Forms

10.9 Vector Analysis

Chapter 11

The Lebesgue Theory

11.1 Set Functions

11.2 Construction of the Lebesgue Measure

11.3 Measure Spaces

11.4 Measurable Functions

11.5 Simple Functions

11.6 Integration

11.7 Comparison with the Riemann Integral

11.8 Integration of Complex Functions

11.9 Functions of Class $\mathcal{L}^2$

Appendix: Construction of $\mathbb{R}$

Bibliography

- [1] Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill International Editions.
- [2] Kenneth Ross, *Elementary Analysis*, Springer Nature.
- [3] Steven Douglass, *Introduction to Mathematical Analysis*, Pearson.
- [4] Terence Tao, *Analysis I*, Springer Nature.